Beyond Line Charts: Advanced Visualization Techniques for Network Telemetry

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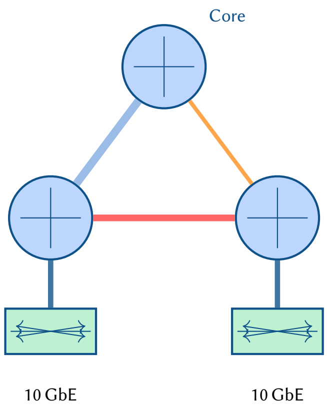


Abstract. Network telemetry systems generate vast streams of time-series metrics, yet operators overwhelmingly rely on simple line charts and threshold-based dashboards to interpret this data. This report surveys fifteen advanced visualization techniques—from topology-embedded overlays and chord diagrams to flame graphs, treemaps, parallel coordinates, and geospatial overlays—and demonstrates each with concrete TikZ renderings. Grounded in Cleveland and McGill's graphical perception hierarchy and Munzner's task–data–idiom framework, we argue that matching the visualization to the analytical question (flow symmetry, latency decomposition, hierarchical allocation) dramatically improves situational awareness and reduces mean time to resolution. We review existing NOC dashboard platforms and identify systematic gaps in their visual vocabulary.

Contents

gNMI	gRPC Network Management Interface	2
IPFIX	IP Flow Information Export	4
CPU	Central Processing Unit	4
QoS	Quality of Service	9
ECMP	Equal-Cost Multi-Path	13
NOC	Network Operations Centre	2
SLA	Service Level Agreement	10
KPI	Key Performance Indicator	6
WAN	Wide Area Network	10
RTT	Round-Trip Time	9
BGP	Border Gateway Protocol	27
ML	Machine Learning	12
PCA	Principal Component Analysis	12
YANG	Yet Another Next Generation	11
SVG	Scalable Vector Graphics	30

1 Introduction

Crucially, telemetry is not just for performance monitoring; it is the mechanism by which we verify intent. By treating the Network Source of Truth (NSoT) and the compiled Plan Topology as an **Anchor of Reality**, we can continuously map live telemetry against the mathematical design. This allows operators to automatically calculate **Intent Drift**—the exact delta between the whiteboard architecture and the operational state.

Modern network infrastructure generates telemetry at unprecedented scale. Streaming protocols such as gRPC Network Management Interface (gNMI) [1] and model-driven telemetry via OpenConfig [2] deliver per-interface counters at sub-second intervals. Yet the default visualization remains the time-series line chart [3], a technique essentially unchanged since the strip-chart recorder.

Insight: *Operators drown in metrics but starve for insight—the bottleneck is visualization, not collection.*

The mismatch between data richness and visual poverty has real operational cost. When an Network Operations Centre (NOC) engineer investigates a congestion event, a wall of line charts forces sequential scanning across dozens of interfaces. Traffic asymmetries, hierarchical bottlenecks, and correlated anomalies hide in plain sight.

This report surveys advanced visualization techniques drawn from information visualization research [4, 5] and adapts them to network telemetry. Each section presents a technique, discusses its analytical strengths, and provides a TikZ reference rendering. fig. 1 establishes the baseline: a raw topology annotated with numeric metrics, the typical starting point that we aim to surpass.

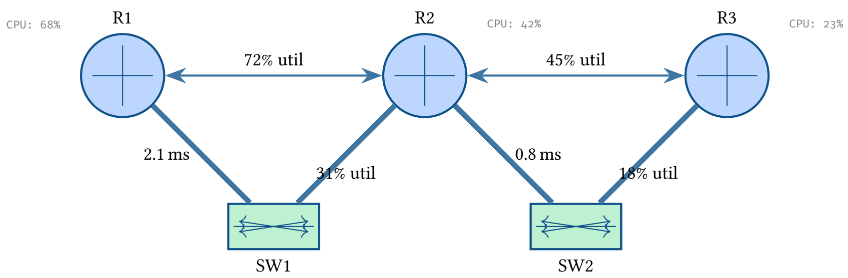


Figure 1. Baseline: raw numeric annotations on a topology diagram. Operators must mentally parse and compare values—a cognitive burden that scales poorly.

2 Perceptual and Mathematical Foundations

2.1 Mathematical Foundation of a Telemetry Stream

Before surveying specific techniques and their perceptual mappings, we must formally define the data structure of network telemetry.

Let $G = (V, E)$ be a directed graph representing the network topology. A telemetry stream S is a sequence of tuples (t, u, attr, v), where $t \in \mathbb{R}$ is the timestamp, $u \in V \cup E$ is the network element, attr is the metric (e.g., ‘if-octets‘), and $v \in \mathbb{R}^d$ is the value vector.

The visualization task is to find a mapping $M : S \to \mathcal{A}$, where $\mathcal{A}$ is a set of visual attributes (position, color, size, opacity, time/animation) that maximizes the information transfer $I(S; \mathcal{A})$ while minimizing the cognitive load $C(\mathcal{A})$.

2.2 Perceptual Hierarchy

With the telemetry stream defined, we establish the perceptual science that guides visualization design. Cleveland and McGill's landmark study [6] ranked graphical encodings by the accuracy with which humans decode them: position along a common scale is most accurate, followed by length, angle, area, and finally color saturation. This hierarchy explains why line charts (position encoding) persist—they are perceptually optimal for small-n trend comparison.

Insight: *Position beats color: a bar chart conveys magnitude more accurately than a heat map, but a heat map scales to thousands of cells where bars cannot.*

Munzner's nested model [7] provides a complementary framework: visualization design operates at four levels—domain situation, data and task abstraction, visual encoding and interaction idiom, and algorithm. Failures at any level cascade downward. A NOC dashboard that presents the wrong *abstraction* (e.g., per-packet counts when the operator needs flow-level aggregates) cannot be rescued by better color palettes.

Design Rule 1: Encoding Selection Principle

Select the visual encoding whose perceptual accuracy matches the precision required by the analytical task. Use position for exact comparison ($n < 20$), length or area for approximate ranking ($20 < n < 200$), and color for pattern detection ($n > 200$).

Ware's pre-attentive processing theory [8] further explains why topology overlays (section 4) are effective: color hue, orientation, and size differences are detected in under 250 ms regardless of the number of distractors. Anomaly highlighting exploits this channel—a single red node among blue ones "pops" without serial scanning.

fig. 2 illustrates the Cleveland–McGill hierarchy applied to network telemetry encodings.

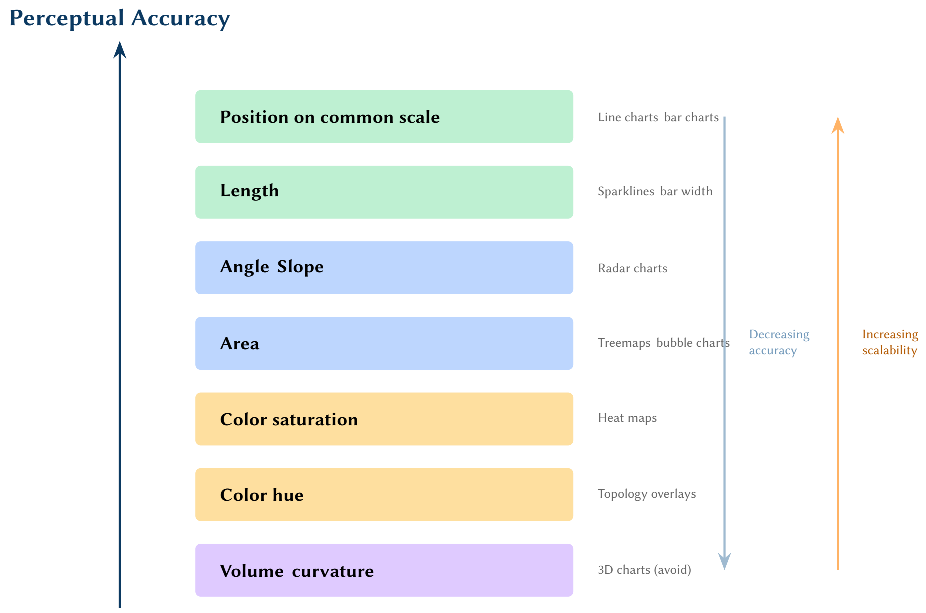


Figure 2. Cleveland–McGill perceptual accuracy hierarchy mapped to network telemetry techniques. Higher encodings are more accurate but less scalable; lower encodings handle larger data sets.

Shneiderman's visual information-seeking mantra—"Overview first, zoom and filter, then details on demand" [9]—provides the interaction model. A telemetry dashboard should open with a topology overview (spatial context), allow filtering by time range or device group, and reveal per-interface detail on click.

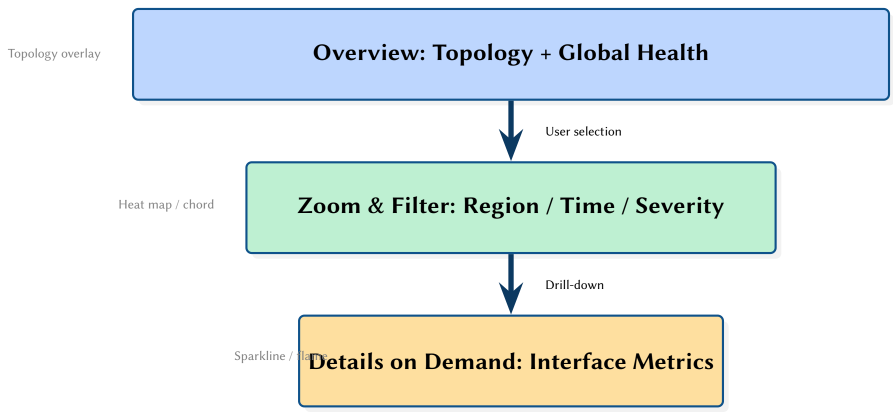


Figure 3. Shneiderman's visual information-seeking mantra applied to network telemetry dashboards. Each level maps to appropriate visualization idioms.

3 Related Work

3.1 Commercial NOC Platforms

The dominant NOC visualization platforms—Grafana [3], LibreNMS [10], Observium [11], and PRTG [12]—share a common visual vocabulary: time-series line charts, gauge widgets, and tabular device lists. Grafana's plugin ecosystem extends this to heat map panels, but the default "rows of line charts" layout remains pervasive.

Insight: *Commercial platforms optimize for generality, not for the specific cognitive tasks of network troubleshooting.*

Kentik [13] represents a more advanced approach, offering Sankey-style flow visualizations and geographic overlays, but remains a proprietary platform without extensible rendering.

3.2 Network-Specific Visualization Research

NetSage [14] pioneered geographic flow visualization for research networks, mapping IP Flow Information Export (IPFIX) data onto world maps with arc width proportional to volume. The Science DMZ concept [15] motivated purpose-built dashboards that go beyond generic monitoring.

Batfish [16] and Minesweeper [17] focus on configuration analysis rather than live telemetry, but their topology visualizations demonstrate the value of embedding analytical results directly into network diagrams.

3.3 Information Visualization Foundations

The techniques surveyed here draw on a rich body of visualization research. Chord diagrams derive from Circos [18], originally developed for genomics. Treemaps were introduced by Shneiderman [19] and refined by Bruls et al. [20]. Flame graphs were invented by Gregg [21] for CPU profiling and have since been applied to distributed tracing. Horizon charts were studied by Heer et al. [22], who showed they support accurate comparison in one-third the vertical space of standard area charts. Parallel coordinates, introduced by Inselberg [23], remain the primary technique for visualizing high-dimensional data without dimensionality reduction.

3.4 Advanced Telemetry Paradigms and Data Science

As networks transition to programmable data planes, traditional polling is being replaced by *In-Band Network Telemetry (INT)* and eBPF-based kernel observability. Recent work such as SF-INT [24] and SATS [25] demonstrate the viability of per-packet and application-level telemetry. These high-resolution, high-volume streams demand visualizations that move beyond scalar line charts.

Furthermore, analyzing these massive telemetry streams increasingly relies on complex systems and signal processing techniques. Topological Data Analysis (TDA) [26] and manifold learning algorithms like UMAP [27] provide mathematical frameworks for projecting hyper-dimensional routing and telemetry data into interpretable geometric spaces. Raster-based density estimation tools like Datashader [28] have proven that massive point-cloud datasets can be visualized as continuous fields without browser-crashing vector overplotting.

> **Gap Analysis**
>
> Existing NOC platforms lack systematic support for: (1) topology-embedded metric encoding, (2) hierarchical decomposition views (treemaps, flame graphs), (3) multi-dimensional health profiles mapped to geographic or geometric structures (UMAP, Datashading), and (4) flow relationship diagrams (chord, Sankey). This report addresses each gap by translating advanced data science into operational "eye candy."

CORE VISUALIZATION IDIOMS

4 Topology-Embedded Overlays

The most natural evolution of the annotated topology is to encode metrics visually: link color for utilization, node size for load, opacity for health. This approach is standard in tools like Weathermap, but rarely implemented with the full expressive range of modern rendering [5].

Insight: *Encoding metrics as visual properties exploits pre-attentive perception—anomalies "pop" without reading numbers.*

fig. 4 demonstrates gradient-colored links (green to red by utilization) and proportionally sized router nodes (by Central Processing Unit (CPU) load).

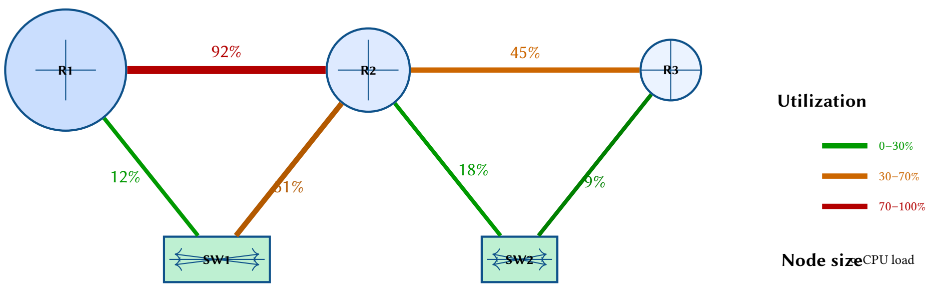


Figure 4. Topology overlay: link color encodes utilization, node diameter encodes CPU load. The R1–R2 link immediately stands out as a congestion point.

Small multiples—placing the same topology side by side at different time instants—reveal temporal evolution without animation [4]. fig. 5 shows two snapshots.

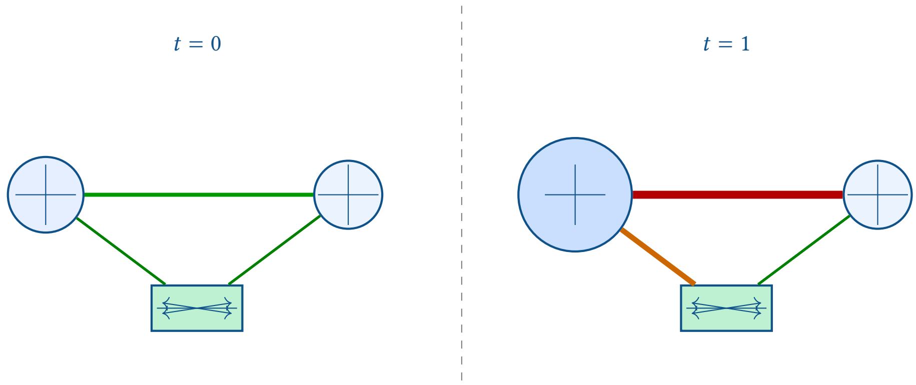


Figure 5. Small multiples: the same topology at two time instants reveals that the left router transitioned from idle to heavily loaded.

5 Heat Maps and Matrix Views

When the number of source–destination pairs or interface–time combinations grows large, individual line charts become impractical. Heat maps encode magnitude as color intensity across a two-dimensional grid.

5.1 Device-to-Device Latency Matrix

Insight: Heat maps exploit the human ability to detect color clusters—a capability line charts cannot offer when $n > 20$.

fig. 6 shows a device-to-device latency matrix. Diagonal entries (self) are zero; off-diagonal entries are colored from green (low latency) through yellow to red (high latency).

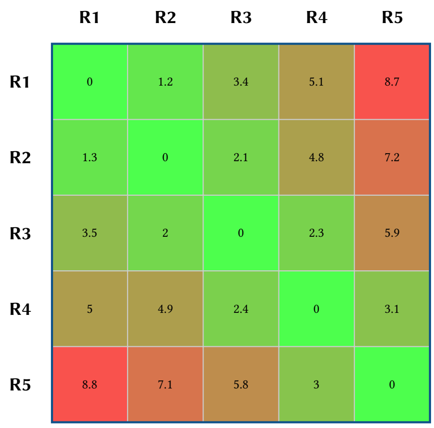


Figure 6. Device-to-device latency matrix (ms). Warm colors highlight high-latency pairs; the R1–R5 path is clearly the worst performer.

5.2 Time-of-Day Utilization Heat Map

fig. 7 maps interface utilization across a 24-hour period. Each row is an interface; each column is an hour. This view immediately reveals diurnal patterns and sustained hot spots.

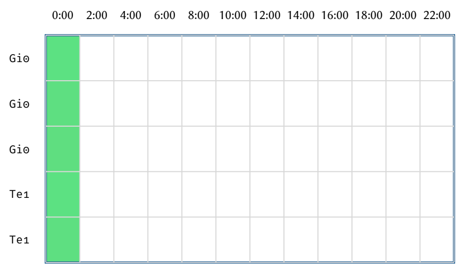


Figure 7. Time-of-day utilization heat map. Gi0/3 shows sustained high utilization during business hours (08:00–20:00).

6 Chord Diagrams

Chord diagrams display flow relationships between entities arranged on a circle, with arc width proportional to traffic volume. They excel at revealing asymmetric flows and dominant communication pairs [18].

Insight: Chord diagrams answer "who talks to whom and how much?"—the quintessential traffic engineering question.

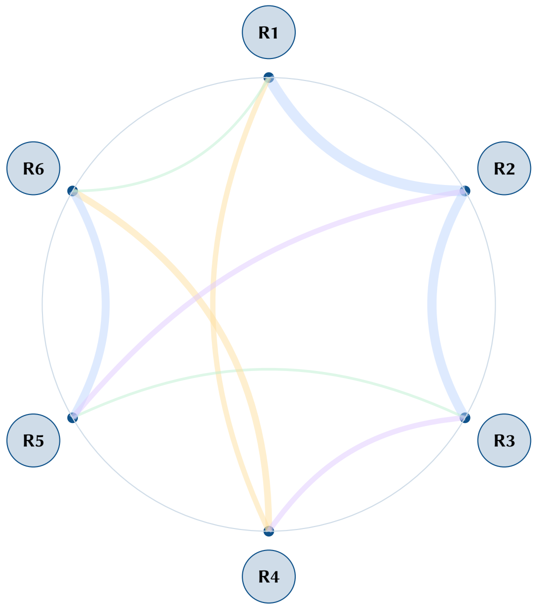


Figure 8. Chord diagram of inter-router traffic flows. Arc width is proportional to volume; the R1–R2 and R2–R3 paths carry the heaviest load.

7 Sankey Diagrams

Sankey diagrams visualize flow through a multi-stage system, with band width proportional to volume. For network telemetry, they map naturally to traffic flowing through network tiers: ingress, aggregation, core, and egress.

Insight:
Sankey diagrams make bandwidth conservation visible—if flow width narrows, traffic is being dropped or shaped.

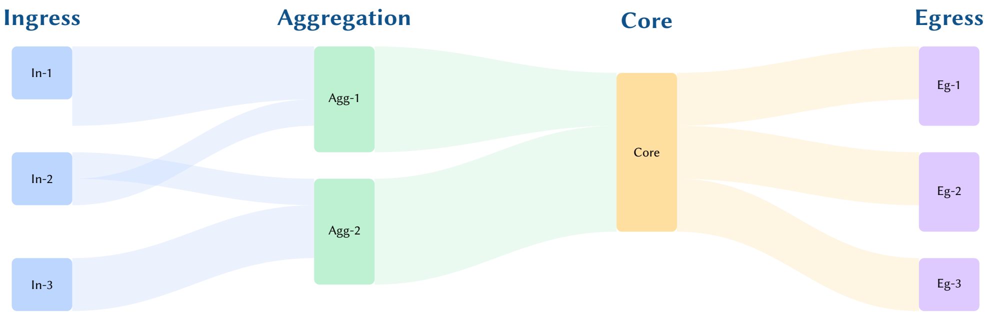


Figure 9. Sankey diagram of traffic flow through network tiers. Band width is proportional to throughput; narrowing bands indicate traffic shaping or filtering at tier boundaries.

8 Radar Charts

Radar (spider) charts plot multiple metrics on radial axes emanating from a common center. They are ideal for comparing the health profile of a device across several Key Performance Indicators (KPIs) simultaneously.

Insight:
Radar charts instantly reveal lopsided health profiles—a "spiky" polygon means one metric dominates, signalling imbalance.

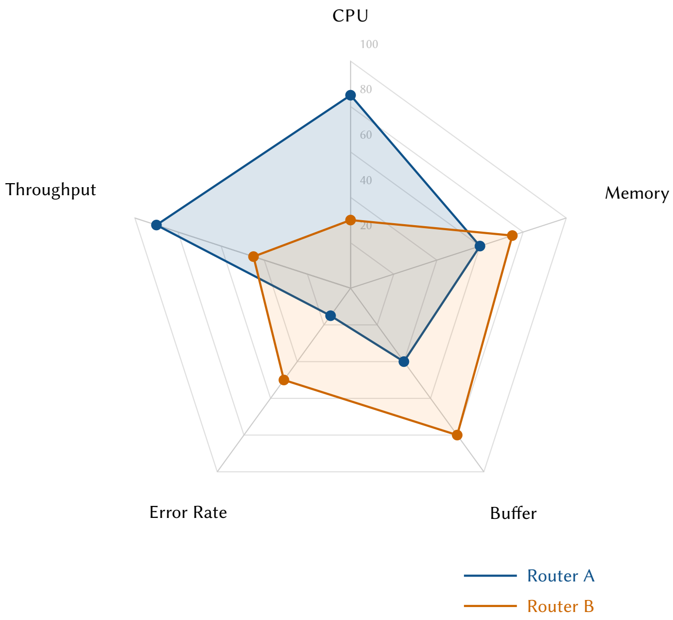


Figure 10. Radar chart comparing two routers. Router A is CPU- and throughput-heavy; Router B has high buffer usage and elevated error rates, suggesting a queuing issue.

9 Sparklines and Horizon Charts

When monitoring many interfaces simultaneously, individual plots must be compressed to fit. Sparklines—small, word-sized inline graphics [4]—convey trend without axes or labels.

Insight: *Sparklines trade precision for density: 50 interfaces on one screen instead of 5.*

9.1 Sparkline Strips

fig. 11 shows sparkline strips for six interfaces. Each strip conveys the last hour's utilization trend in a single row.

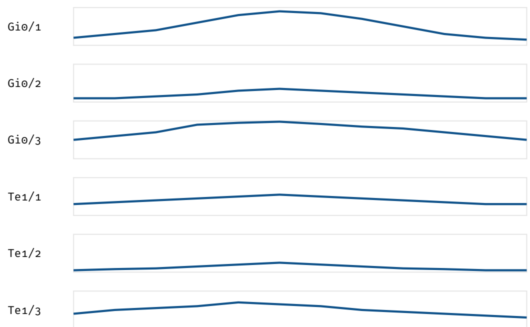


Figure 11. Sparkline strips for six interfaces. Gi0/3 shows sustained high utilization; Te1/2 remains lightly loaded throughout.

9.2 Horizon Charts

Horizon charts fold the vertical axis into layered colored bands, doubling or tripling the data density of a standard area chart. Values are divided into bands; higher bands overlay lower ones with increased color intensity. fig. 45 illustrates this folding for a single interface.

Figure 12. Horizon chart: three intensity bands fold the y-axis into one-third of the vertical space. Darker regions indicate higher utilization.

10 Treemaps

Treemaps encode hierarchical data as nested rectangles, with area proportional to a quantitative variable. For network telemetry, the hierarchy site → device → interface maps naturally to bandwidth allocation [20].

Insight: *Treemaps reveal allocation imbalance at every level of the hierarchy simultaneously.*

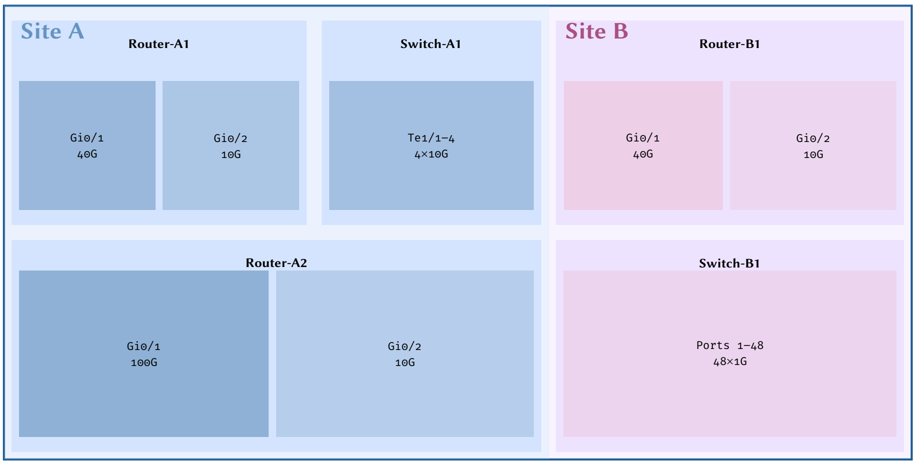


Figure 13. Treemap of bandwidth allocation: site → device → interface. Area is proportional to allocated bandwidth. Site A's 100G uplink on Router-A2 dominates the view.

11 Flame Graphs

Flame graphs, popularized for CPU profiling [21], decompose total latency into hierarchical contributions. Applied to packet processing, they show where time is spent as a packet traverses the forwarding pipeline.

Insight:
Flame graphs turn latency from a single number into a narrative: "60% of delay is in QoS classification, not forwarding lookup."

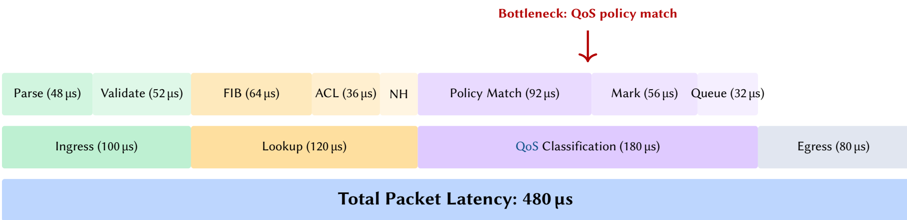


Figure 14. Flame graph of packet processing latency. QoS classification consumes 37.5% of total latency, with policy matching as the dominant sub-stage.

12 Hive Plots and Structural Fabric Analysis

In dense, highly connected fabrics like Spine-Leaf datacenters, standard force-directed layouts often collapse into unreadable "hairballs". Hive plots solve this by mapping network elements onto fixed, radially distributed axes based on their structural role (e.g., Spine, Leaf A, Leaf B), and drawing curved links between them.

Insight:
Hive plots expose structural rule violations. Because the layout is deterministic rather than physics-based, an anomalous "East-West" link directly between two Leaf nodes (which should only communicate via a Spine) instantly stands out as a geometric violation.

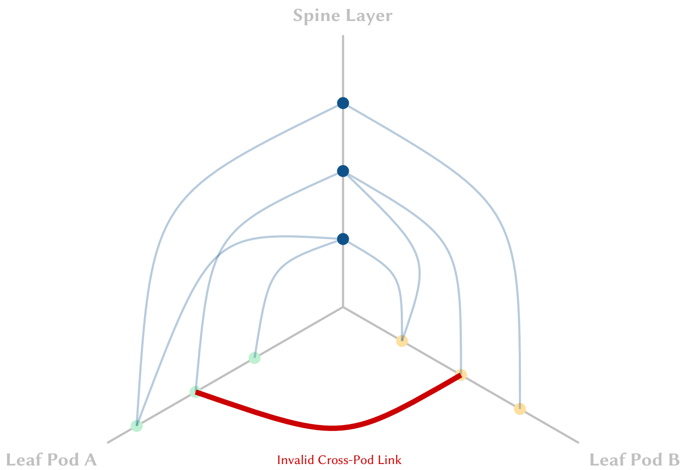


Figure 15. Hive Plot of a Spine-Leaf Fabric. The rigid, rules-based positioning of nodes on semantic axes makes the illegal direct connection between Pod A and Pod B glaringly obvious, a detail that would be lost in the chaos of a force-directed web.

13 Force-Directed Layouts and Anomaly Highlighting

Force-directed (spring-model) graph layouts position nodes such that heavily-connected nodes cluster together and lightly-connected nodes drift apart. When edge "spring constant" is set proportional to traffic volume, the layout itself becomes a visualization of traffic affinity [29]. Anomalous nodes—those violating expected patterns—can be highlighted with color or shape changes.

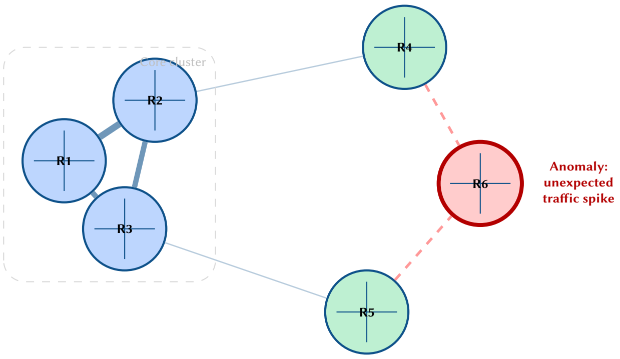


Figure 16. Force-directed layout with anomaly highlighting. Core routers cluster by traffic affinity; R6 (red) exhibits unexpected traffic patterns and is flagged for investigation.

14 Parallel Coordinates

Parallel coordinates [23] plot each variable on its own vertical axis; each data point is a polyline connecting its values across all axes. For network telemetry, each line represents a device and axes represent metrics (CPU, memory, throughput, error rate, Round-Trip Time (RTT)).

The technique scales to dozens of dimensions—far beyond what radar charts support—and readily highlights outliers as lines that diverge from the main bundle [7].

Insight:
Parallel coordinates reveal correlations between metrics: crossing lines indicate inverse relationships; parallel bundles indicate positive correlation.

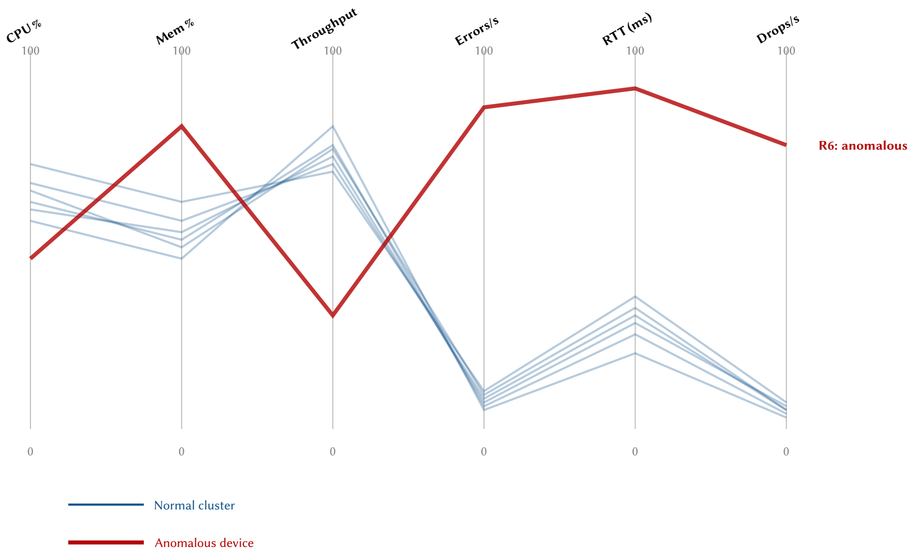


Figure 17. Parallel coordinates for six device metrics. The normal devices (blue) form a coherent bundle; R6 (red) diverges with high error rate, RTT, and packet drops.

15 Stream Graphs

Stream graphs (a variant of stacked area charts with a displaced baseline) show how the composition of aggregate traffic evolves over time [30]. Each "stream" represents a protocol, application class, or Quality of Service (QoS) marking, and the total height at any point encodes the aggregate throughput.

The organic shape also makes visual outliers (sudden protocol bursts) highly salient.

Insight:
Stream graphs answer "what changed in the traffic mix?"—a question invisible to per-metric line charts.

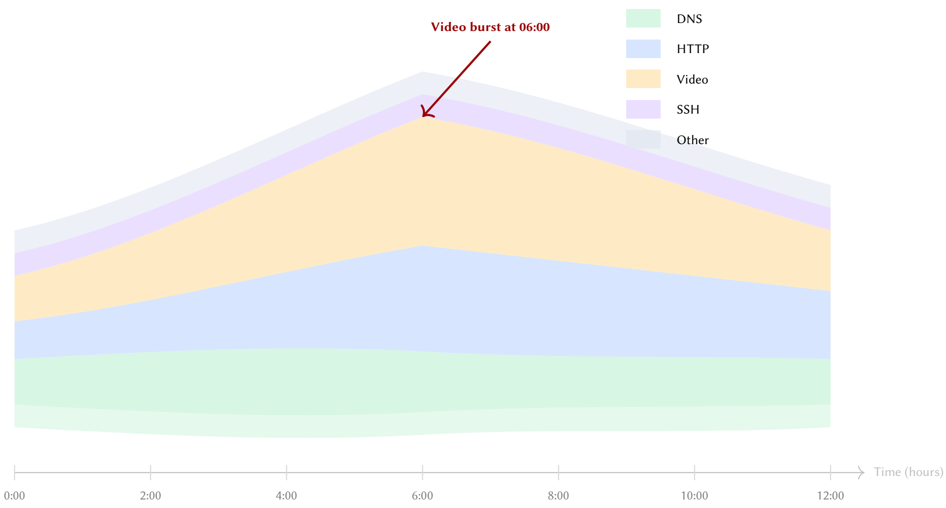


Figure 18. Stream graph of traffic composition over 12 hours. The video stream (amber) dominates mid-day; a burst at 06:00 is immediately visible as asymmetric widening.

16 Waterfall Charts

Waterfall charts decompose a cumulative total into sequential contributions. In network telemetry, they naturally map to *hop-by-hop latency accumulation*: each bar segment represents the delay added by a single network element along a path. Unlike flame graphs (section 11), which decompose processing within a single device, waterfall charts trace latency across the network [29].

Insight: *Waterfall charts turn end-to-end SLA violations into actionable per-hop blame assignments.*

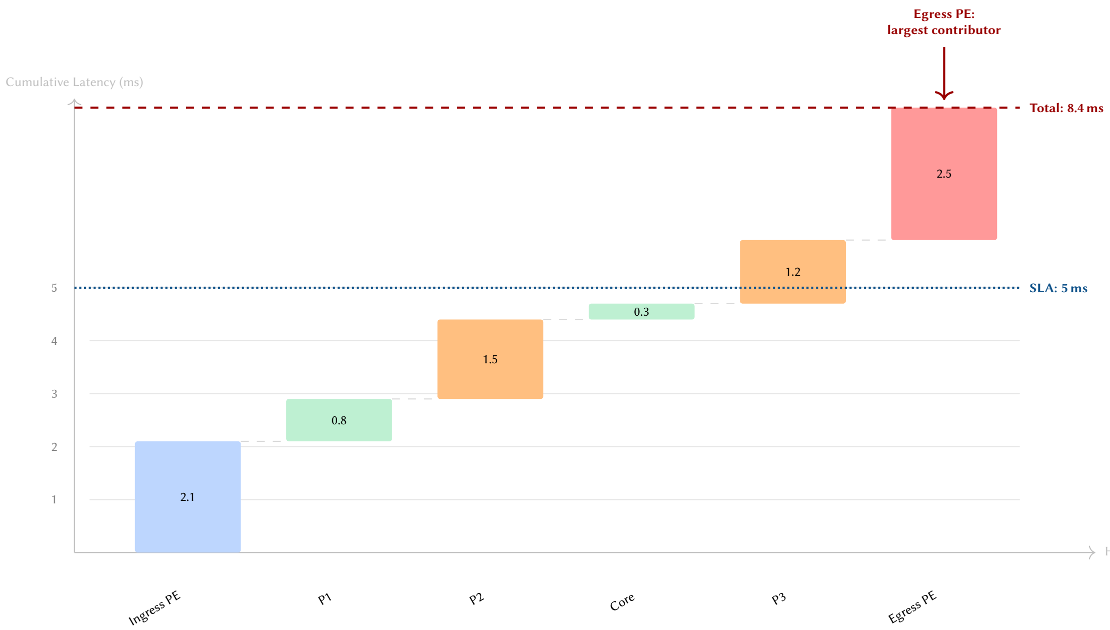


Figure 19. Waterfall chart of hop-by-hop latency accumulation. The egress PE contributes 2.5 ms (30% of the total), pushing the path above the 5 ms SLA threshold.

17 Geospatial Overlays

For Wide Area Network (WAN) telemetry, geographic context is essential. Geospatial overlays plot metrics onto schematic or real maps, with arcs representing inter-site links and visual properties encoding utilization [14].

fig. 20 shows a schematic continental layout with utilization-coded arcs and site health markers.

Insight: *Geographic overlays answer "where is the problem?" before "what is the problem?"—spatial context accelerates triage for distributed networks.*

Telemetry Architecture and Pipeline

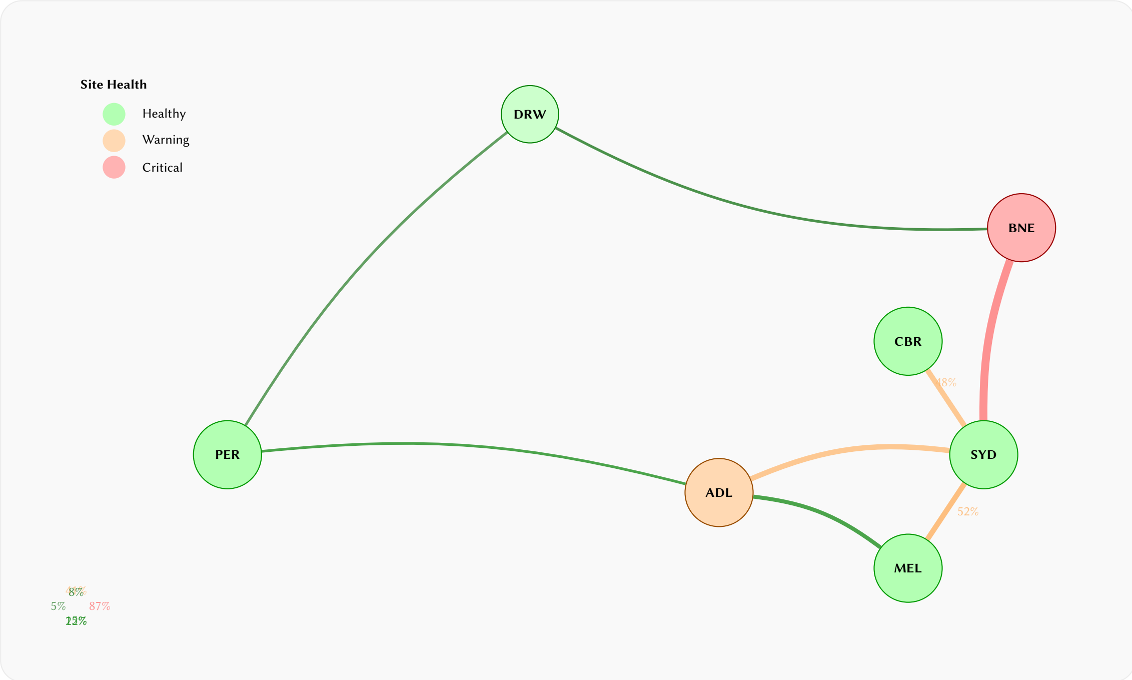
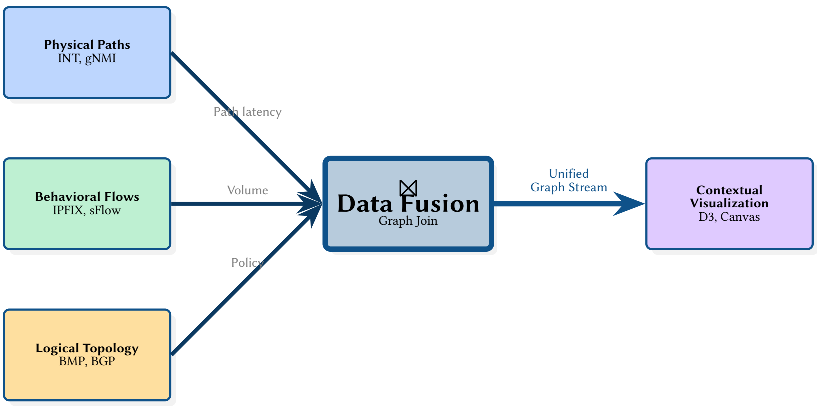


Figure 20. Geospatial overlay of an Australian WAN. Site color indicates health; arc width and color encode link utilization. The SYD–BNE link is near saturation.

18 Input Categorization and Telemetry Data Fusion

Before metrics reach a dashboard, they traverse a multi-stage pipeline: collection, transport, storage, query, and rendering [31, 32]. To unlock advanced visualizations, we must first categorize our input sources. Different protocols provide fundamentally different topological "lenses," and true insight often requires *fusing* these distinct data streams.

- **Model-Driven Telemetry (gNMI/gRPC):** Replaces legacy SNMP with high-frequency, push-based streaming of structured Yet Another Next Generation (YANG) models. Provides sub-second resolution of interface stats, CPU, and optical power.
- **Traffic Flow (IPFIX/sFlow):** Sampled endpoint conversations (Source IP, Dest IP, Port). Provides the behavioral "who is talking to whom" without knowing the physical path.
- **Routing State (BMP/BGP):** Control-plane messages and route advertisements. Provides the logical overlay, reachable prefixes, and intended pathing.
- **Active Probing (TWAMP/ICMP):** Synthetic packets injected to measure exact end-to-end latency, jitter, and packet loss from a user's perspective.
- **In-Band Network Telemetry (INT):** Per-packet metadata. Provides the holy grail: exact pathing, microburst queuing, and per-hop latency simultaneously.



Insight: *The most powerful "eye candy" is born from Data Fusion. Visualizing flow data (IPFIX) overlaid on top of logical routing state (BMP), colored by active probing latency (TWAMP), creates a holistic, multivariate view that no single protocol can provide.*

Figure 21. The Data Fusion Pipeline. Advanced visualization is not merely a rendering problem; it is a data-joining problem. By performing a join (⋈) between physical path data (INT), behavioral flows (IPFIX), and logical routing state (BMP) before the rendering stage, we produce a unified "eye-candy" view that reveals the interplay between intent and reality.

19 Anomaly Detection Feedback Loop

Advanced dashboards do not merely display metrics—they close the loop between automated anomaly detection and human investigation [33]. Machine Learning (ML)-based detectors (autoencoders, isolation forests, Principal Component Analysis (PCA)-based methods) flag statistical outliers, but operators need visual context to confirm and diagnose them.

fig. 22 illustrates this feedback loop as a conceptual architecture.

Insight:
Visualization is the interface between machine detection and human judgement—without it, anomaly alerts become noise.

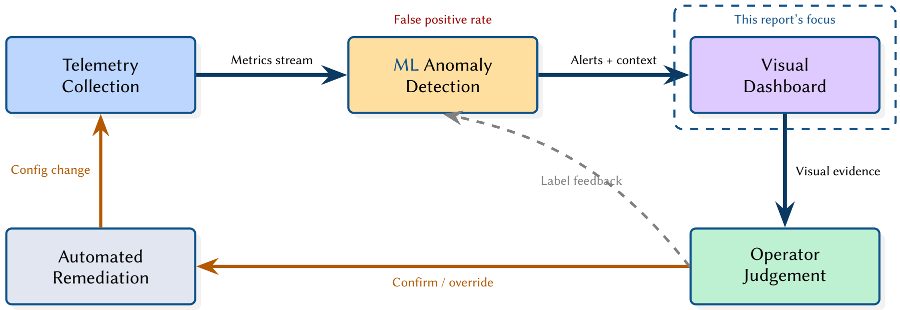


Figure 22. Anomaly detection feedback loop. Visualization (highlighted) bridges machine detection and human judgement, enabling operators to confirm, override, or label anomalies for model retraining.

20 Multi-Resolution Time-Series

Network events span vastly different time scales: micro-bursts last microseconds, congestion episodes span minutes, and capacity trends evolve over months. A single time axis cannot serve all three. Multi-resolution views stack synchronized panels at different temporal granularities, linked by a shared selection brush [9].

Insight:
Multi-resolution views let operators move from "is there a problem this month?" to "what happened at 14:32:07?" without switching tools.

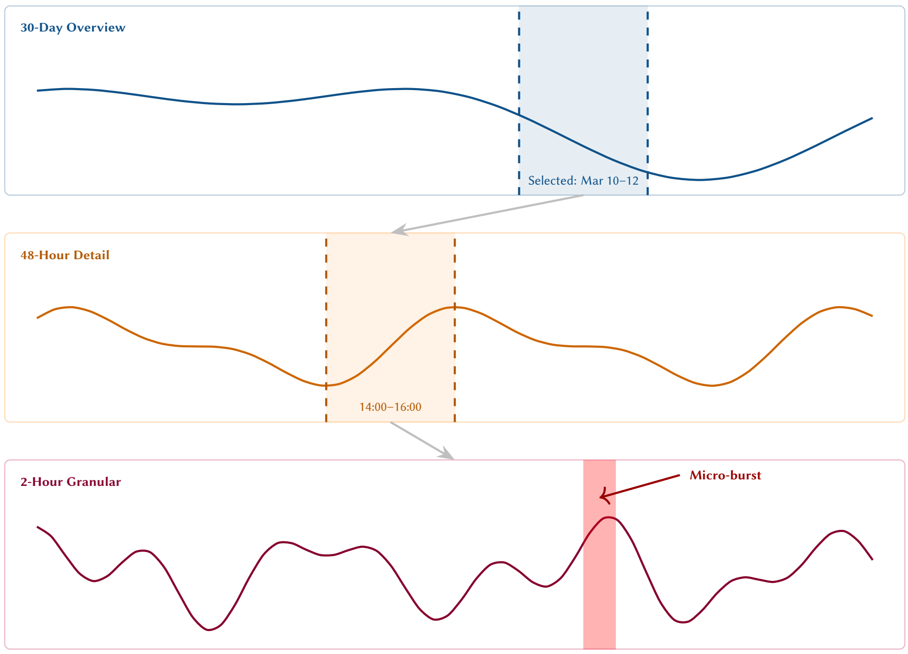


Figure 23. Multi-resolution time-series with linked selection brushes. A monthly overview narrows to a 48-hour window, then to 2-hour granularity where a micro-burst becomes visible.

21 Topology Diff Visualization

Network topologies evolve: links fail, devices are added, routing reconverges. Visualizing the *difference* between two topology snapshots—before and after an event—helps operators understand the scope of change. This is analogous to a "diff" view in version control, applied to graph structure rather than text [16].

Insight:
Topology diffs turn implicit "what changed?" questions into explicit visual answers—added elements in green, removed in red, unchanged in gray.

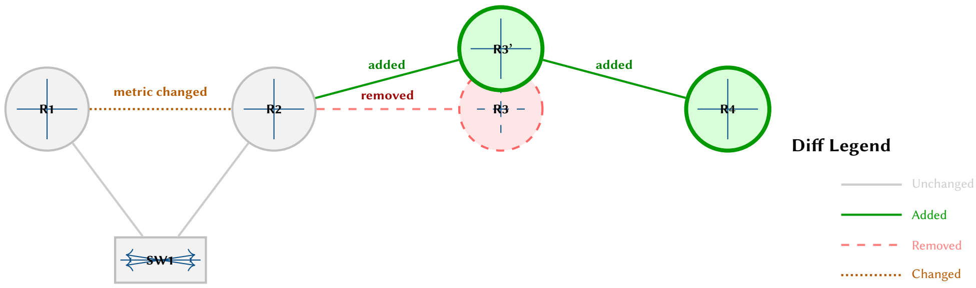


Figure 24. Topology diff visualization. After a maintenance event, R3 was replaced by R3' and R4 was added (green); the old R3 link was removed (red dashed); the R1–R2 link metrics changed (orange dotted).

22 ECMP Path Diversity Visualization

Equal-Cost Multi-Path (ECMP) routing distributes traffic across multiple equal-cost paths, but operators often lack visibility into the actual distribution. Visualizing all available paths with width proportional to observed traffic share reveals load imbalance that simple utilization charts miss.

Insight:
ECMP visualizations answer "is my traffic actually balanced?"—a question that per-link utilization views cannot address without path-level context.

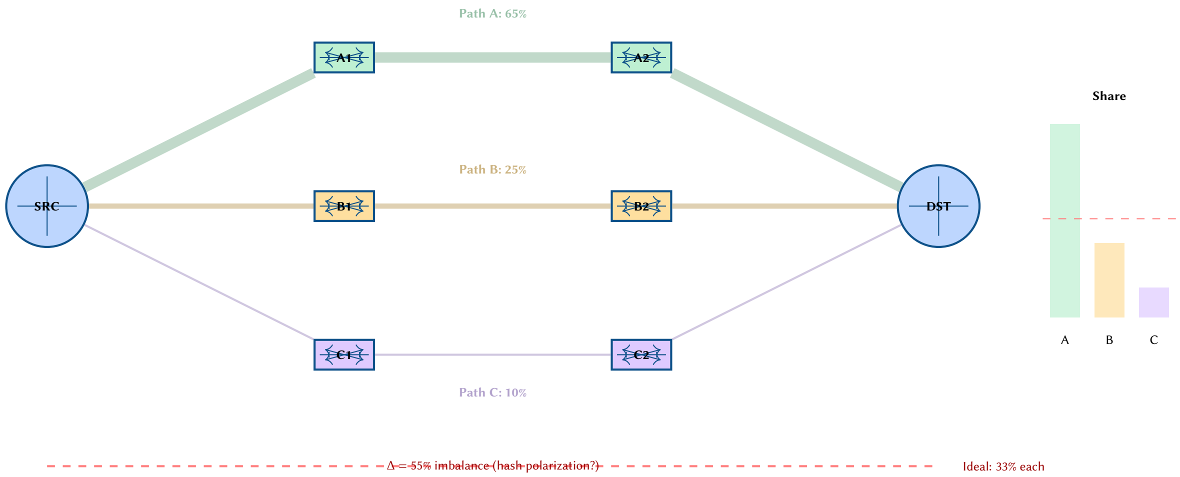


Figure 25. ECMP path diversity visualization. Three equal-cost paths carry 65%, 25%, and 10% of traffic respectively—far from the ideal 33% split, suggesting hash polarization.

23 Telemetry Abstraction Layers

A recurring theme in this report is that different visualization techniques operate at different levels of abstraction: from raw per-interface counters to aggregated flow summaries to topological health assessments. fig. 26 presents this as a conceptual stack, mapping each abstraction level to its appropriate visualization idioms and analytical questions.

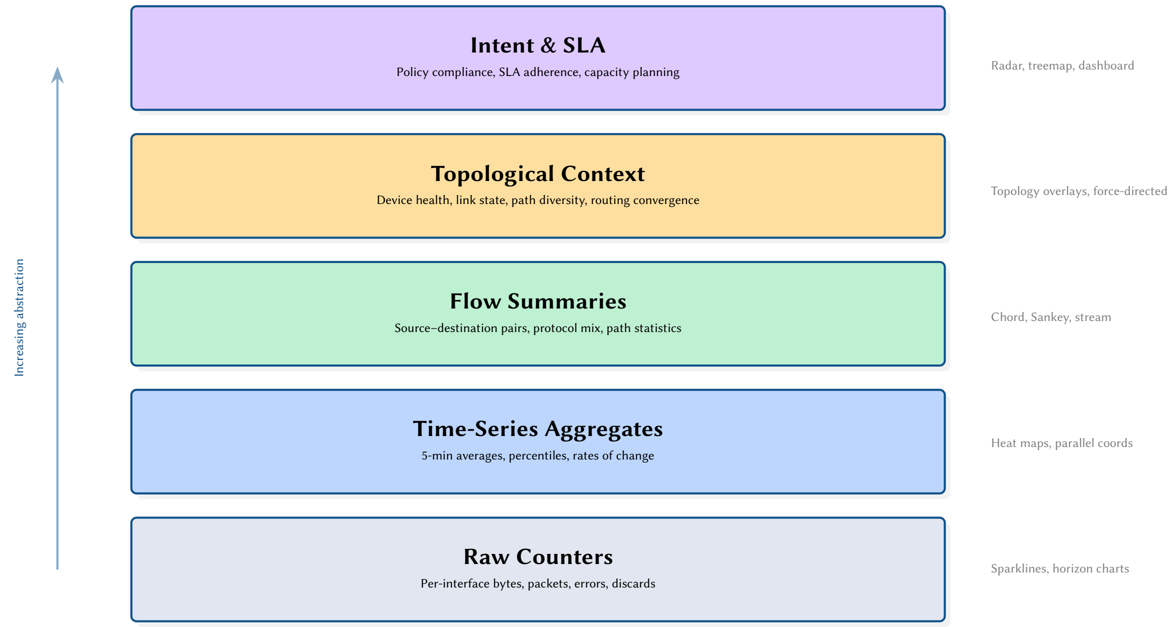


Figure 26. Telemetry abstraction stack. Each layer poses different analytical questions and calls for different visualization idioms. Dashboards should span multiple layers to support the full diagnostic workflow.

24 Dashboard Composition

No single visualization answers every operational question. Effective dashboards compose multiple complementary views, linked by shared time context and interactive filtering [34].

fig. 27 shows a mock dashboard layout combining four of the techniques discussed in this report.

Insight:
Dashboard design is information architecture—choosing which views to juxtapose is as important as choosing the views themselves.

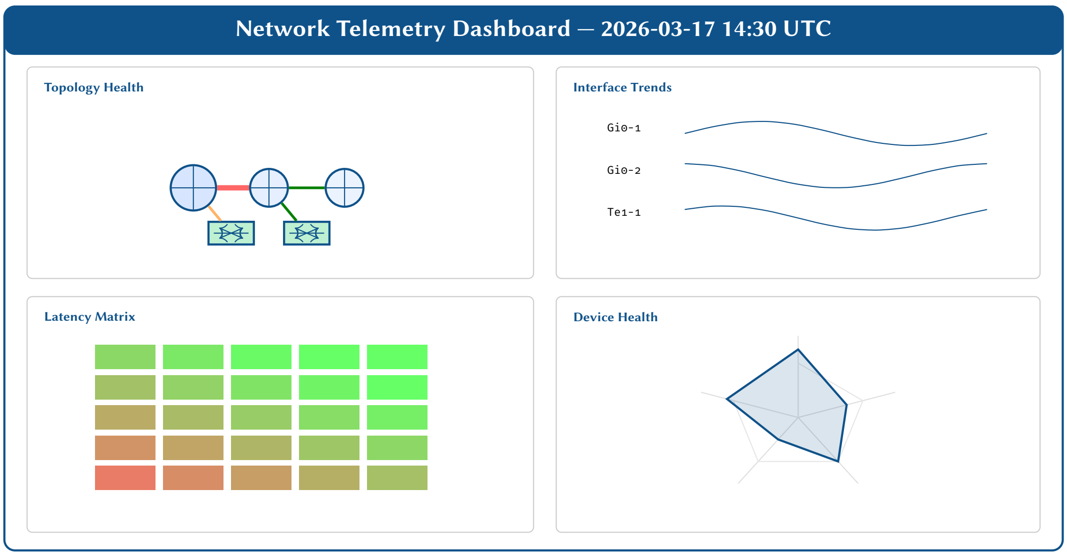


Figure 27. Composite dashboard layout combining topology overlay, sparkline trends, latency matrix, and radar health views. Linked selection and shared time context enable rapid cross-referencing.

Advanced Frontiers and Aesthetics

25 Emerging Frontiers: Sonification, AR, and Topology

The techniques discussed thus far primarily map to spatial and color dimensions on 2D screens. However, modern telemetry platforms are beginning to incorporate techniques from sensory modalities, spatial computing, and complex systems analysis.

25.1 Network Sonification (Acoustic Telemetry)

The human auditory system is exceptionally skilled at detecting subtle changes in rhythm, pitch, and harmony, often reacting faster to a broken rhythm than to a changing pixel. *Sonification* maps network telemetry to audio signals.

Instead of static diagrams, imagine a continuous data stream where:

Insight:
If a datacenter is running smoothly, it generates a low, rhythmic ambient "hum" (e.g., a baseline heartbeat mapping BGP updates to a bass drum, and TCP flows to a white-noise hiss). When a

- **Pitch** maps to Interface Utilization.
- **Tempo/Rhythm** maps to Packet Rate.
- **Timbre/Distortion** maps to Error Rate or Packet Loss.
- **Spatial Audio (Left/Right)** maps to Geographic Origin.

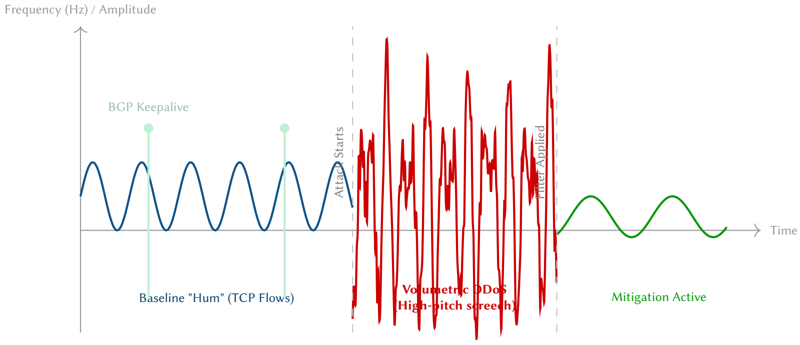
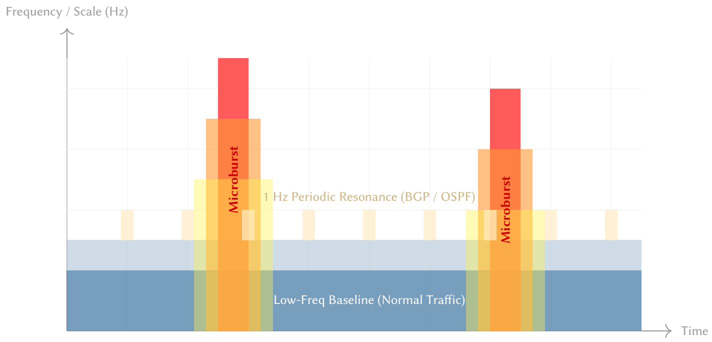


Figure 28. Visual Representation of Acoustic Telemetry (Sonification). The visual translates an auditory experience: a calm, low-frequency hum periodically punctuated by a BGP "heartbeat," suddenly transforming into a high-amplitude, high-frequency "screech" during a volumetric attack.

25.2 Wavelet Spectrograms (Frequency-Domain Telemetry)

Standard time-series line charts perfectly capture low-frequency trends (like a slow diurnal increase in traffic). However, they systematically hide high-frequency anomalies, like a 5-millisecond TCP window synchronization event that causes a microburst. By applying a Continuous Wavelet Transform (CWT), we convert a 1D time-series into a 2D *Spectrogram* heatmap.



Insight:
A Spectrogram allows operators to see the resonance of the network. A slow baseline shift is a bright color at the bottom (low frequency). A dangerous microburst that causes queue drops appears as a sudden, bright vertical "flame" shooting up into the high-frequency spectrum, revealing anomalies that averages mathematically erase.

Figure 29. Wavelet Spectrogram of Link Utilization. Time is on the X-axis, and Frequency is on the Y-axis. The solid blue base shows normal background traffic. The two bright red vertical "flames" instantly expose high-frequency microbursts that would be completely flattened and invisible on a standard 1-minute averaged MRTG/SNMP line chart.

25.3 Augmented Reality (AR) Datacenter Overlays

In a physical datacenter, a network engineer diagnosing a faulty Spine switch must cross-reference the physical cables in the rack with a logical dashboard on their laptop. Spatial computing (AR headsets) eliminates this gap by anchoring the telemetry directly onto the physical hardware.

25.4 Topological Data Analysis (TDA)

As networks become hyper-dimensional (e.g., a BGP table with millions of routes overlapping with MPLS labels and VXLAN tunnels), even our most advanced 3D visual methods fail. Topological Data Analysis (TDA) is a mathematical framework that studies the "shape" of high-dimensional data by extracting its persistent topological features (connected components, holes, and voids) [26].

Insight:
Through an AR headset, the engineer looks at a physical switch rack and sees glowing "holographic" bars hovering above each port. A dead optic is instantly identifiable as a red glowing port, and the operator can see a virtual 3D [illegible] tracing the physical path of the cable through the ceiling to its destination.

Insight:
Instead of [illegible] through 100,000 overlapping BGP routes, TDA computes the abstract "shape" of the routing table. If a malicious actor injects a massive block of fake routes, the mathematical "shape" of the network tears, creating a new topological "hole." The [illegible] simply flags the sudden appearance of this hole, turning an unreadable 100,000-line table into a single geometric alert.

25.5 Graph Signal Processing (GSP)

In GSP, the network topology $G = (V, E)$ forms the spatial domain, and telemetry metrics (e.g., link utilization) are treated as a signal $\mathbf{x} \in \mathbb{R}^{|V|}$ living on the vertices or edges [35]. By computing the Graph Fourier Transform (GFT) using the eigenvectors of the graph Laplacian $\mathbf{L}$, we can analyze telemetry in the *graph frequency domain*.

Insight:
GSP allows us to detect "high-frequency" telemetry anomalies—situations where adjacent routers have wildly different states (e.g., routing loops or asymmetric ECMP hashing) that are invisible when looking at devices in isolation.

25.6 QoS Ternary Operating Points

When analyzing Quality of Service (QoS) queue drops, standard line charts fail because QoS is a zero-sum game: an increase in Expedited Forwarding (EF) bandwidth forces a decrease in Best Effort (BE) or Assured Forwarding (AF). Ternary plots map this three-way trade-off onto a 2-simplex (a triangle).

Insight:
QoS is a zero-sum game. The ternary plot shows the "operating point" of a router within a triangle. If one class expands, it must

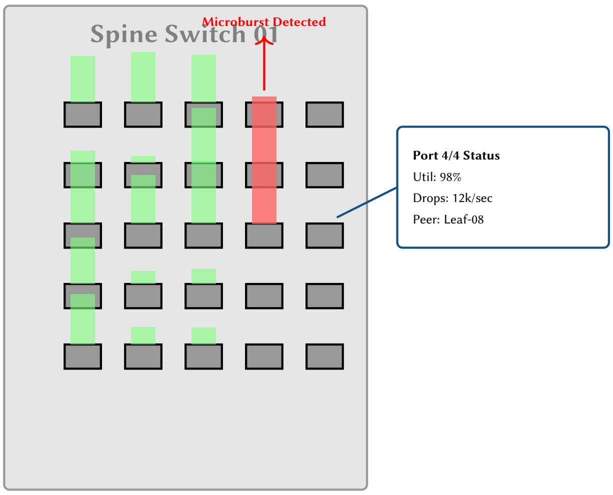


Figure 30. AR Datacenter Overlay. A concept for spatial computing where live telemetry (port utilization bars and diagnostic panels) is projected as a hologram directly over the physical network hardware, eliminating the need to cross-reference a separate dashboard.

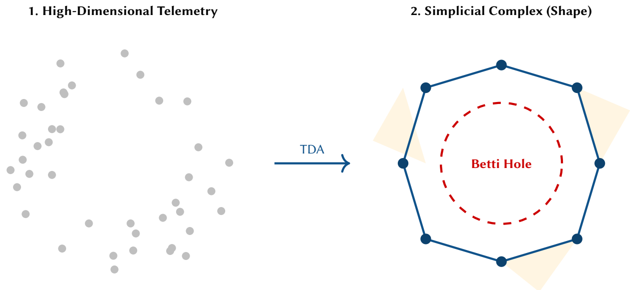


Figure 31. Topological Data Analysis (TDA). The algorithm takes a dense, unreadable point cloud of routing telemetry (left) and computes its mathematical structure (right). The emergence of the topological "hole" (a non-trivial 1-cycle) mathematically proves an anomaly, such as a routing loop or a severed path, regardless of how many dimensions the original data possessed.

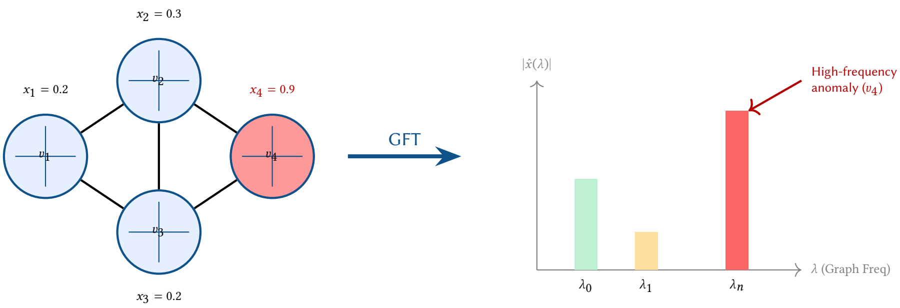


Figure 32. Graph Signal Processing (GSP) maps spatial telemetry into graph frequencies. The sudden jump in metric at v_4 registers as a massive high-frequency spike in the GFT domain.

25.7 Digital Twins and 3D Isometric Projection

As network operations evolve from passive monitoring to predictive simulation, the concept of the *Digital Twin* has emerged: a real-time, synchronized virtual replica of the physical network. Visualizing a digital twin requires moving beyond 2D flat maps to separate the physical underlay from the logical overlay (e.g., BGP, VPN, or SDN tunnels).

25.8 In-Band Network Telemetry (INT) Metadata Snowball

Traditional telemetry relies on external polling or asynchronous streaming. *In-Band Network Telemetry (INT)* embeds telemetry instructions and the resulting data directly into the user-plane packets as they transit the network. To visualize INT, we must invert the perspective: the visualization follows the packet's journey rather than the router's state.

Insight: *3D Isometric projection allows us to stack layers vertically. A routing anomaly can be traced by dropping a plumb line from a congested logical BGP peering down to the physical fiber cut that caused it.*

Insight: *Visualizing INT requires tracking the "metadata snowball" effect. As a packet traverses switches, its header grows, accumulating per-hop latency and queue depth data, which is then extracted at the edge.*

25.9 eBPF and Kernel Datapath Observability

With the rise of Kubernetes and Cloud-Native Network Functions (CNFs), the traditional router boundaries have dissolved into the Linux kernel. *eBPF (Extended Berkeley Packet Filter)* allows tracing packets at microscopic

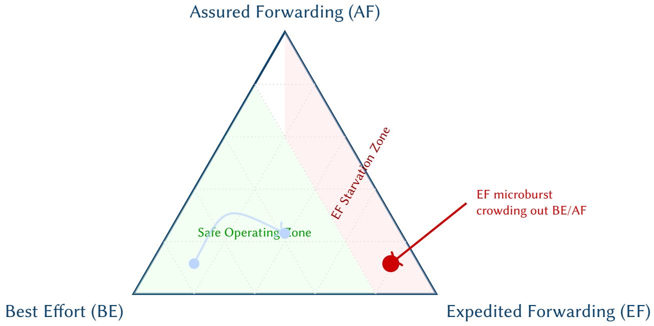


Figure 33. Ternary plot of QoS allocation. Because the sum of BE, AF, and EF must equal 100% of the interface bandwidth, the router's state is a single point in the triangle.

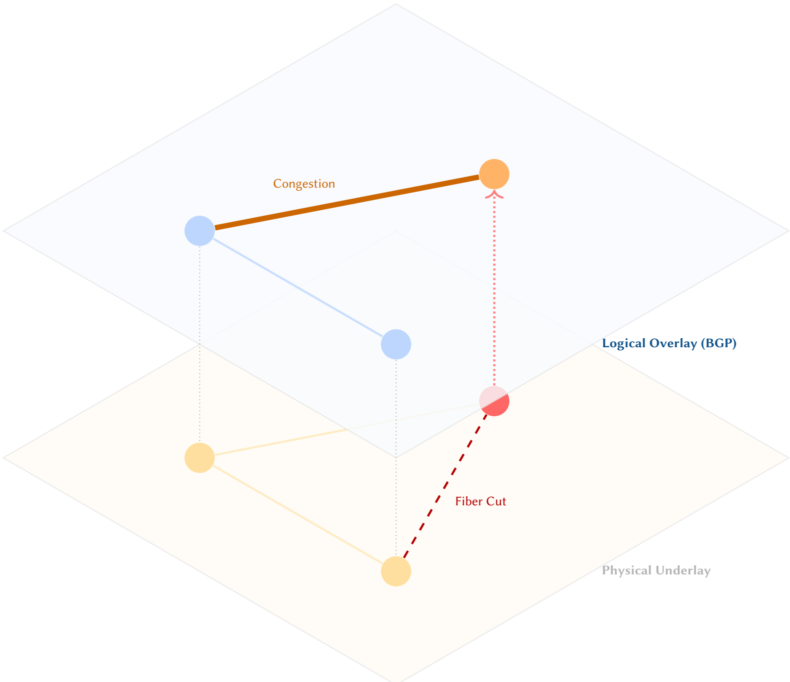


Figure 34. 3D Isometric projection of a network Digital Twin. A physical fiber cut in the underlay cascades vertically, causing a routing shift and subsequent congestion in the logical overlay. Plumb lines visually correlate the layers.

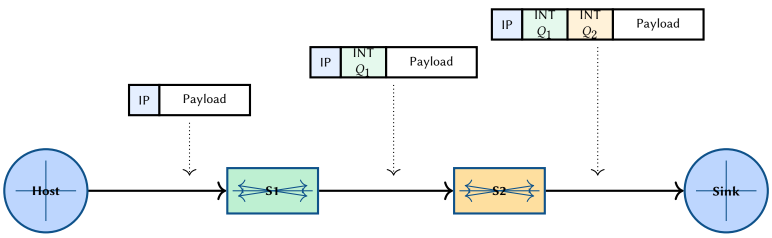


Figure 35. In-Band Network Telemetry (INT) pipeline. The telemetry metadata (e.g., queue depths Q) "snowballs" as the packet transits the data plane, providing perfect hop-by-hop visibility without synthetic probes.

kernel hooks (XDP, TC, Socket Layer). Visualizing this requires abandoning the "node-and-link" topology entirely, replacing it with a kernel stack traversal diagram.

25.10 Explainable AI (XAI) Saliency Maps

As Autonomous Networks increasingly rely on Graph Neural Networks (GNNs) for root-cause analysis, a critical challenge is trust. Operators will not approve an AI-driven remediation if they cannot understand its reasoning. Visualizing AI requires *saliency maps*—highlighting the specific nodes, links, and metrics that contributed most heavily to the model's conclusion.

Insight:
eBPF visualization exposes the invisible networking within a single host. It answers: "Did the packet drop in the physical NIC, the iptables firewall, or the application socket?"

Insight:
XAI visualizes the model's attention, *not just the network state. If an AI flags a node as the root cause, the visualization must highlight the precise gradient path (the "why") that led to that classification.*

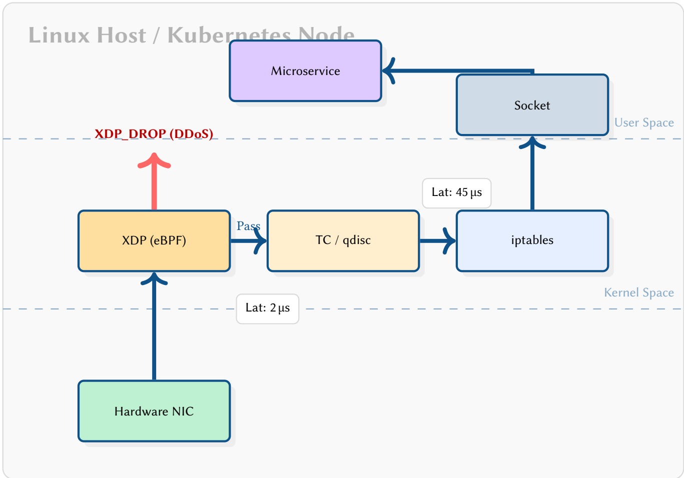


Figure 36. eBPF Datapath Observability. Visualizing the kernel networking stack reveals that a DDoS flow was efficiently dropped at the XDP (eXpress Data Path) layer before it could consume host CPU resources.

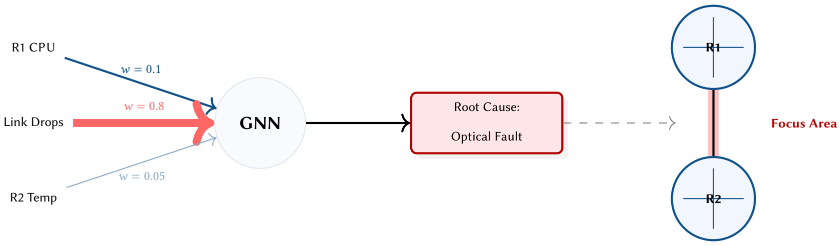


Figure 37. XAI Saliency Map. The visualization shows that the Graph Neural Network heavily weighted ($w = 0.8$) the "Link Drops" feature over CPU or temperature. The resulting projection highlights the specific link in the topology, building operator trust.

26 Tufte-Inspired Telemetry Aesthetics

Edward Tufte's principles of information design—maximizing the data-ink ratio, utilizing micro/macro readings, and escaping "flatland" (low-dimensional plotting)—offer a radical departure from standard NOC dashboards. Instead of using gauges and boxy widgets, we can use the ink itself to represent the physical and temporal realities of the network. The following designs strip away "chartjunk" to focus purely on analytical narrative.

26.1 The Minard Flow: Visualizing a BGP Route Leak

Inspired by Charles Joseph Minard's famous 1869 map of Napoleon's Russian campaign, this diagram maps six dimensions simultaneously: logical path, physical distance (x-axis), traffic volume (band thickness), latency (color saturation), packet loss (decaying width), and temporal progression (read left to right).

26.2 Phase-Space Trajectories: Escaping the Time-Series

Standard line charts isolate metrics, hiding their continuous interaction. A *phase-space trajectory* plots two metrics against each other over time, revealing the *shape* of the router's state as an orbital path. Tufte's "range frames" (axes drawn only over the range of the data) are used to further reduce chartjunk.

26.3 Nightingale Rose Charts for Diurnal Rhythms

Networks possess deep, rhythmic circadian cycles (e.g., diurnal waking and sleeping patterns of users). Plotting these on a linear X-axis arbitrarily cuts the cycle at midnight. A Nightingale Rose (or Coxcomb chart) wraps the time axis into a 24-hour polar clock face, stacking protocols by radius.

26.4 Alluvial Diagrams for Policy and Routing Decisions

Firewalls, Access Control Lists (ACLs), and BGP routing tables are often treated as invisible "black boxes" where packets disappear. An Alluvial Diagram (a categorical variation of a Sankey diagram) explicitly traces the flow of packets as they intersect with policy rules, dropping or forwarding them into distinct categories.

26.5 Micro/Macro Geographic Glyph Maps

Tufte argues against removing detail to achieve clarity; instead, he advocates for "Micro/Macro readings" where dense, localized data is embedded directly into broad overviews. Rather than painting a city node a simple "red" or "green", we can overlay a multivariate *Glyph* (like a mini radar chart) onto the map.

Insight:
By removing boxes and node borders, the data is the architecture. A BGP route leak becomes immediately visually apparent as a massive "hemorrhage" of ink draining away from the intended topological path.

Insight:
A healthy router orbits in a tight, dense knot of low delay and low buffer utilization. A microburst appears as a sudden, lasso-like excursion into the danger zone, returning to the knot as the queue drains.

Insight:
A healthy network on a Nightingale Rose looks like a symmetric, blooming flower. An anomaly, such as a volumetric DDoS attack or a massive flash crowd, instantly breaks the geometric symmetry of the "bloom," creating a jagged spike that triggers pre-attentive recognition far faster than a linear spike.

Insight:
Alluvial diagrams expose the unexposed. By plotting the sequence of categorical decisions (Ingress → QoS Class → Firewall Policy → Drop/Pass), an operator can visually see exactly which policy rule is indiscriminately "eating" valid Expedited Forwarding (EF) traffic.

Insight:
The viewer perceives the continental topology at the macro-level, but can lean in to read the specific multivariate degradation profile (e.g., Latency vs. CPU) at any individual POP without interacting with the

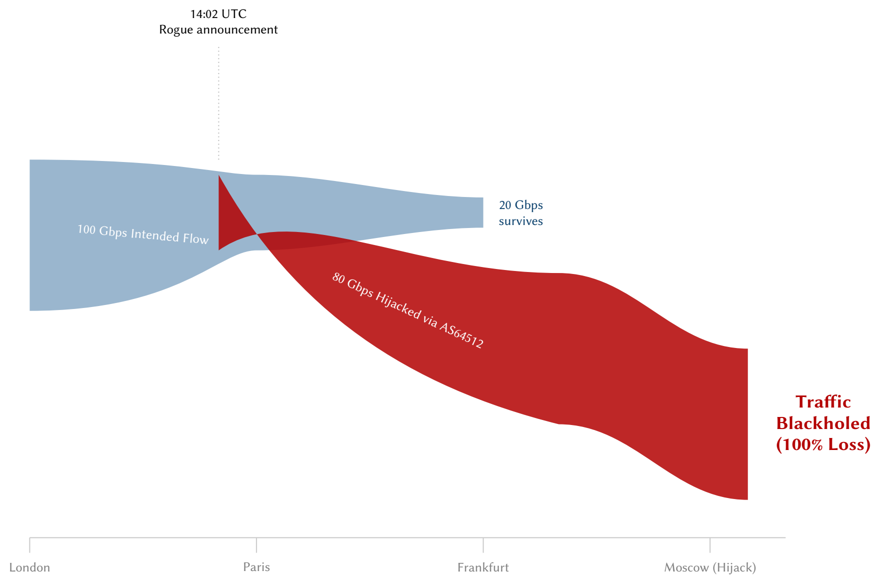


Figure 38. A Minard-style flow map of a BGP route leak. The thickness of the band represents bandwidth, while the divergence vividly illustrates the hijacking of traffic. The data-ink ratio is maximized by eliminating traditional chart bounding boxes.

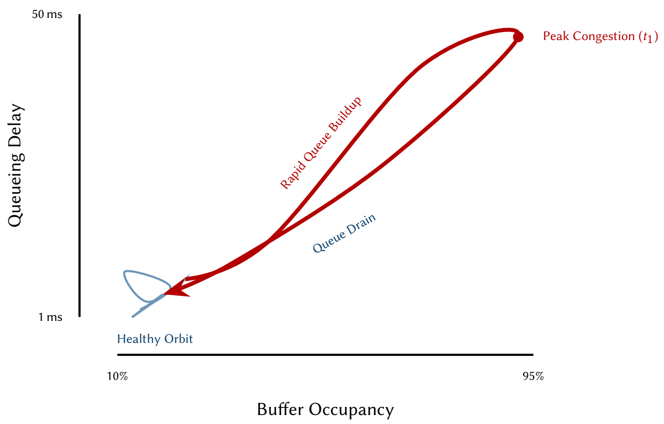


Figure 39. Phase-Space Trajectory. Rather than plotting metrics against time on an x-axis, time is represented by the continuous line itself. The shape of the loop instantly characterizes the severity and duration of the microburst.

26.6 High-Density Sparkline Matrices

Tufte invented sparklines: "data-intense, design-simple, word-sized graphics." By embedding micro-topologies and metric sparklines directly into a tabular layout, we achieve massive data density. The operator's eye can scan dozens of network paths for anomalies without opening a single drill-down menu or changing context.

26.7 Ridgeline Distributions: Exposing the Long Tail

Network operators are often deceived by averages. A standard line chart plotting "average latency" every 5 minutes completely destroys the reality of jitter and micro-congestion. *Ridgeline plots* (or joyplots) solve this by stacking the full Probability Density Function (PDF) of the metric at various time intervals.

26.8 Horizon Graphs: Extreme Data Density

Modern NOC dashboards struggle with scale. Monitoring the utilization of a 48-port switch typically requires 48 separate line charts, pushing most data below the fold. *Horizon Graphs* solve this by slicing the Y-axis into uniform bands (e.g., 0-33%, 33-66%, 66-100%) and overlaying them in the same physical space using progressively darker color saturation.

26.9 Small Multiples: Diagnosing Hash Polarization

In Clos (Spine-Leaf) architectures, Equal-Cost Multi-Path (ECMP) relies on entropy hashing to distribute flows evenly. When this breaks (e.g., due to elephant flows or misconfigured hash tuples), the resulting polarization is nearly impossible to spot on a global throughput gauge. Tufte's principle of *Small Multiples*—repeating the same graphical design structure across a matrix of variables—turns the human eye into an incredibly fast anomaly detection engine.

Insight:
In a ridgeline plot, the volume of the curve is constant. A healthy network exhibits tall, narrow spikes (low jitter). As congestion sets in, the peak flattens and spills to the right, vividly exposing the "long tail" of 99th percentile packet delays that ruin application performance.

Insight:
A Horizon Graph compresses vertical space by up to 75% without losing the exact numerical magnitude of peaks. This allows dozens of high-resolution interfaces to be stacked in the space of a single traditional chart.

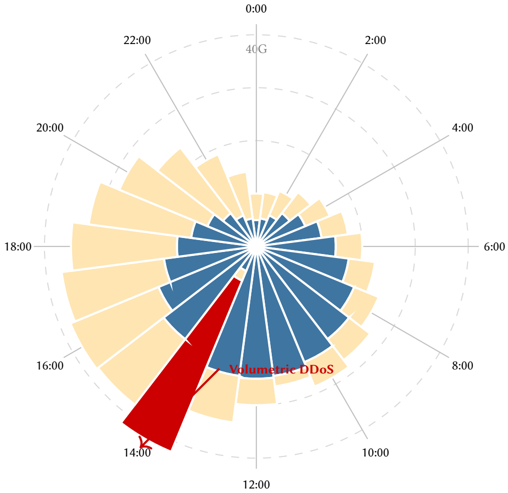


Figure 40. Nightingale Rose Chart of 24-hour Traffic. The inner blue layer represents web traffic; the outer orange layer represents video (peaking in the evening). The sudden red spike at 14:00 shatters the visual symmetry, revealing a high-volume anomaly.

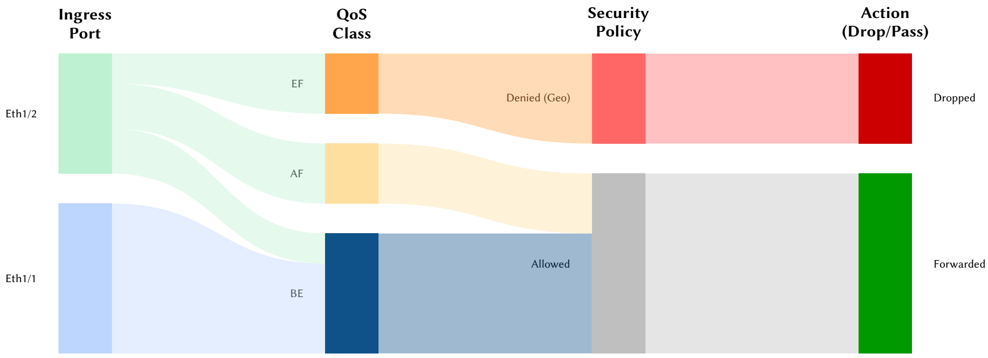


Figure 41. Alluvial Diagram for Policy Tracing. The categorical flow makes it visibly obvious that all Expedited Forwarding (EF) traffic originating from Eth1/2 is intersecting a "Geo Deny" policy and being dropped.

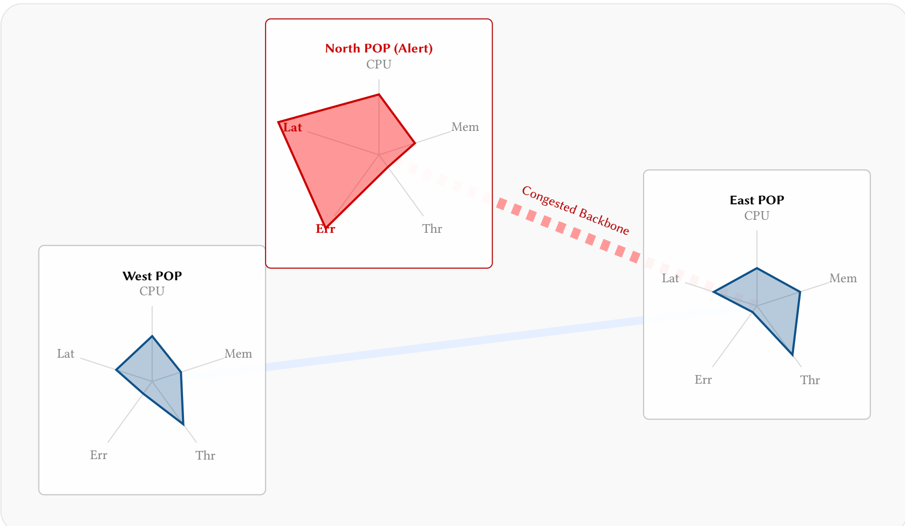


Figure 42. Micro/Macro Geographic Glyph Map. A traditional geographic overlay is supplemented with dense multivariate glyphs at the nodes. The operator observes the regional macro-topology while simultaneously reading the exact micro-metrics causing the North POP alert (Error Rate and Latency spikes).

Critical Path	Micro-Topology	Bandwidth (24h)	Latency (24h)	Loss
London → NYC				0%
Tokyo → SGP				**1.2%**
Sydney → LAX				0%

Figure 43. Sparkline Matrix. Inline micro-topologies provide structural context, while word-sized sparklines reveal 24-hour trends without the distraction of coordinate axes. The red dot highlights a localized latency spike correlated with a reroute in the micro-topology.

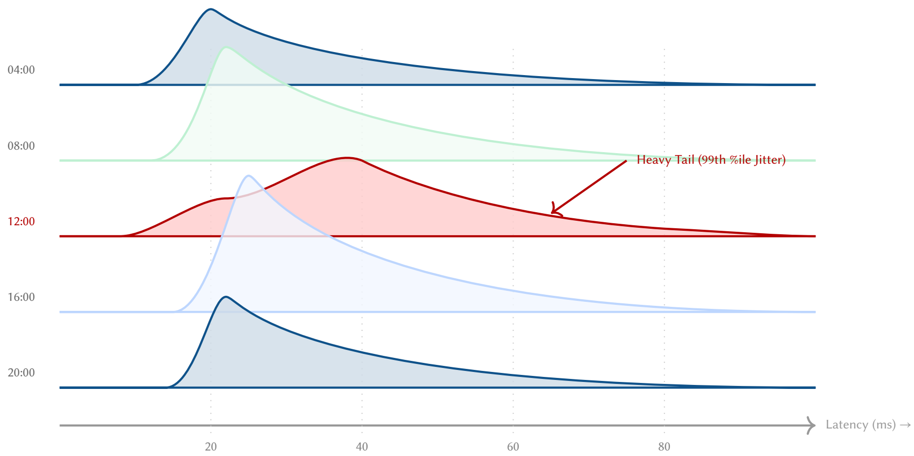


Figure 44. Ridgeline Plot of Latency Distributions. Traditional systems plot the arithmetic mean, hiding the reality of the network. By plotting the full probability density over time, the severe jitter and long tail of packet delays at 12:00 becomes undeniable.

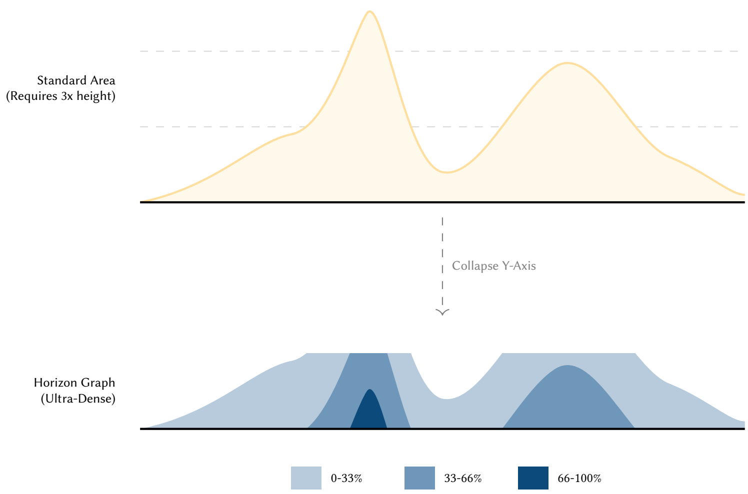


Figure 45. Horizon Graph Construction. A standard area chart (top) wastes vertical space. The Horizon Graph (bottom) slices the Y-axis into bands and overlays them using progressively darker colors. The dark blue peak at $X = 3$ instantly signals a high-magnitude event in a fraction of the space.

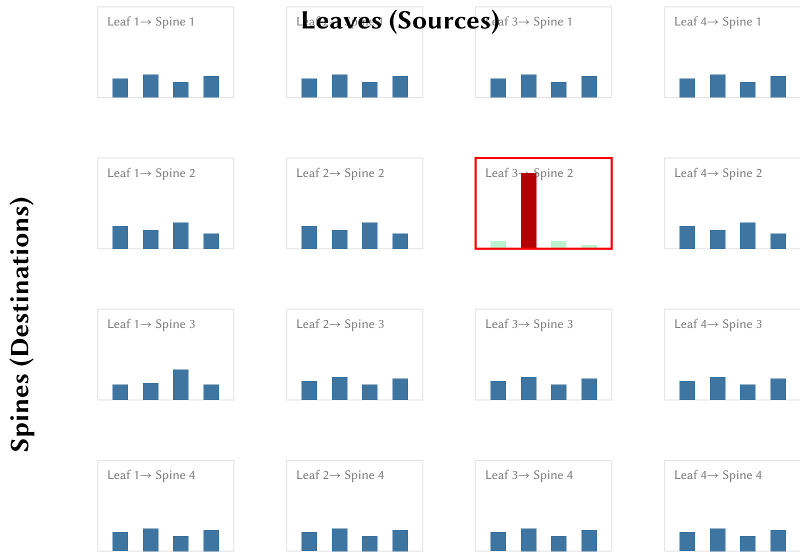


Figure 46. Small Multiples for ECMP Distribution. By repeating a simple micro-bar-chart across a 4×4 grid, the visual system acts as a parallel processor. The eye instantly detects the single aberrant hashing polarization (Leaf 3 to Spine 2) breaking the symmetry of the fabric.

27 Ultra-Scale Telemetry: Pixels and Continuous Fields

When the telemetry scale reaches millions of endpoints—such as individual customer IP addresses or IoT devices—traditional node-link topologies collapse into visual “hairballs”. To handle ultra-scale data, we must abandon discrete nodes and lines in favor of dense pixel displays, continuous raster fields, and edge bundling [36, 37].

27.1 Hilbert Curve Pixel Maps (Dense Pixel Displays)

The IPv4 address space is essentially a 1D line of 4.3 billion points. Visualizing this linearly is impossible. A space-filling Hilbert curve maps this 1D space onto a 2D grid while preserving spatial locality—IPs that are numerically close (in the same subnet) remain physically close on the grid.

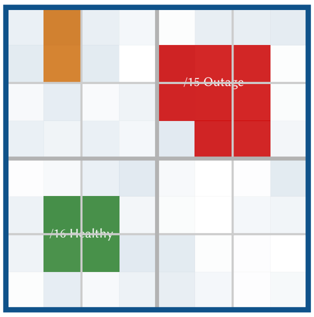


Figure 47. Dense Pixel Display using a Hilbert Curve mapping for a /8 subnet. Each pixel represents a smaller subnet (e.g., /24). The spatial locality of the curve reveals that the outage is confined to a contiguous /15 allocation block, instantly isolating the blast radius.

Insight:
A Hilbert Curve Pixel Map allows operators to see a distributed scanning attack, a BGP route leak, or a localized outage as a coherent geometric shape (e.g., a solid block of red pixels) rather than scattered noise across a dashboard.

27.2 IP Prefix Trees with Hierarchical Edge Bundling

To visualize traffic flows between subnets without creating a chaotic web, we map the IP hierarchy as a radial tree (dendrogram). We then use Hierarchical Edge Bundling [37] to group the flow lines along the branches of the tree. Node sizes at the perimeter can be weighted by customer density or active lease count.

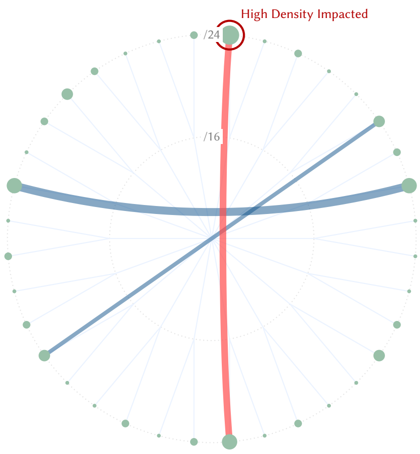


Figure 48. Radial IP Prefix Tree with Hierarchical Edge Bundling. Outer node size encodes customer density. The bundled red line clearly indicates that a major traffic anomaly is hitting one of the most densely populated /24 subnets in the network.

Insight:
Bundling reveals macro-level communication patterns between major network blocks. Weighting the leaf nodes by customer density instantly answers: “Is this BGP flap affecting a densely populated subnet or an empty one?”

27.3 Datashaded Edge Bundling (Kernel Density Fields)

When drawing millions of source-destination IP pairs (e.g., global flow data), even hierarchical edge bundling results in a massive "hairball" of vector lines that crashes the browser’s SVG renderer. Datashading solves this by abandoning vector lines entirely. Instead, each flow is treated as a continuous probability field using Kernel Density Estimation (KDE) and rasterized directly into a pixel grid.

Insight:
Datashading turns a chaotic web of overlapping lines into a glowing, translucent density field. Operators don’t see discrete lines; they see "hot paths" glowing white-hot, instantly revealing the primary arteries of internet traffic without any vector overplotting.

27.4 Rasterized Geo-Density and Isarithmic Maps

When plotting millions of CDN requests or IoT telemetry points on a map, vector nodes suffer from catastrophic overplotting. By rasterizing the data—aggregating it into hexagonal bins or a continuous scalar field (datashading)—we reveal the true shape of the demand. Furthermore, we can treat latency as a topological “elevation” and draw isarithmic (contour) maps.

Insight:
A topographic contour map transforms latency from a spreadsheet number into a terrain feature. Operators can instantly see if a routing failover has turned a “valley” of low latency into a “mountain” of delay.

27.5 Hexagonal Spatial Binning (Hexbins)

While contour maps treat latency as a continuous elevation, visualizing discrete event counts (like security blocks or BGP flaps) across millions of endpoints requires density binning. Standard rectangular grids create

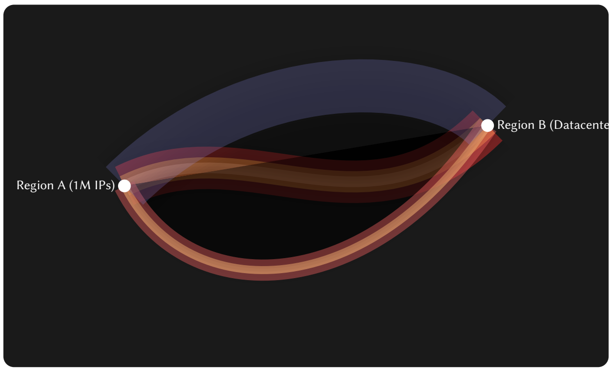


Figure 49. Datashaded Topology (Simulated). Millions of discrete flow records are mathematically aggregated into a continuous raster field. The core routing path glows yellow/white due to massive density, while diffuse, multi-path routing appears as a faint blue/red "cloud" of probability.

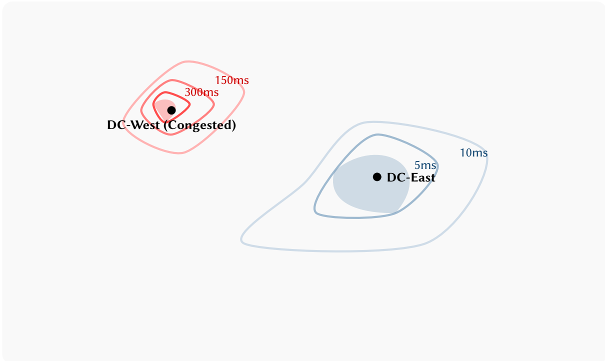


Figure 50. Isarithmic (Contour) Latency Map. DC-East sits in a broad "valley" of low latency. DC-West has experienced a severe degradation, visible as tightly packed contour lines indicating a steep "gradient" of increasing delay for local customers.

visual artifacts and emphasize horizontal/vertical alignment. Hexagonal binning provides a more organic, tightly packed tessellation that better represents radial diffusion.

__Insight:__
Hexbins group scattered, high-volume geographic anomalies into cellular structures. A scanning attack hitting a specific ISP's IP range visually aggregates into a glowing "cell cluster", effectively downsampling the noise without losing the spatial magnitude.

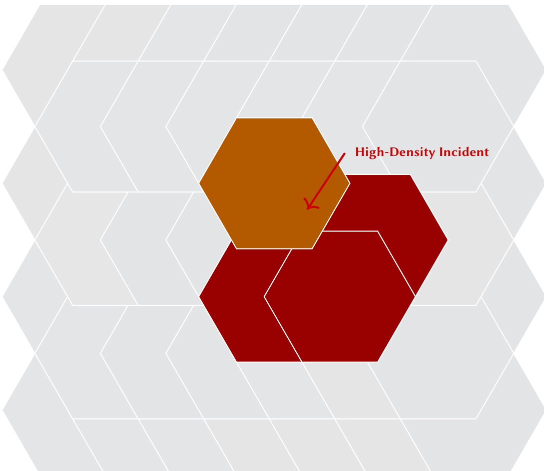


Figure 51. Hexagonal Spatial Binning. Instead of drawing 50,000 overlapping red dots for individual security events, the map space is tiled into hexagons, with color intensity representing event count.

28 Geometric, Distributional, and ML-Assisted Encodings

Presenting raw statistics to network operators often fails to provoke immediate action because standard aggregations (averages, medians, or box plots) hide the true "shape" of the network state. By using geometric, distributional, and machine-learning-assisted encodings, we can transform abstract statistics into tangible visual objects.

28.1 Ridgeline Plots (Joyplots) for Latency Distributions

Operators routinely complain that box plots are difficult to read and hide multimodal data (e.g., a routing flap causing traffic to split evenly between a 10ms path and a 100ms path). A Ridgeline Plot (or Joyplot) solves this by plotting partially overlapping density curves for multiple routers on the same temporal axis.

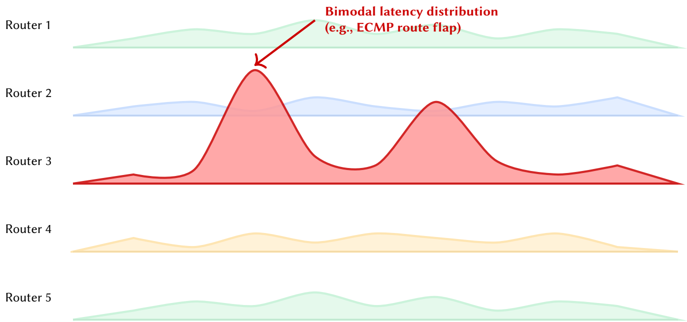


Figure 52. Ridgeline Plot of Latency. The overlapping density curves effectively convey the distribution of delay across five routers. Router 3 exhibits a clear bimodal anomaly (two distinct "peaks"), a vital detail completely lost in an average or median metric.

Insight: *Ridgeline plots create a visual "mountain range" of latency. When a microburst or routing anomaly occurs, an operator doesn't just see a rising line; they see a new "peak" suddenly erupting out of the distribution, providing an intuitive, physical sense of the delay.*

28.2 Dynamic Voronoi Tessellations for Edge Catchment

For Content Delivery Networks (CDNs) or Anycast architectures, operators need to know which datacenter is servicing which users. A Voronoi diagram partitions a map into "catchment areas" based on the closest node (measured by geographic distance or network latency).

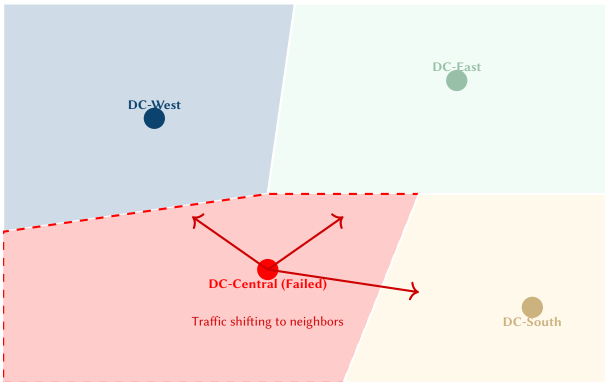


Figure 53. Voronoi Catchment Areas. The diagram visualizes the Anycast latency boundaries. DC-Central has failed (dashed red cell), and the red arrows indicate how the neighboring Voronoi regions will expand to absorb the traffic load.

Insight: *Voronoi tessellations visualize the literal territory of a datacenter. If a node fails or degrades, its Voronoi cell shrinks or collapses entirely, and operators can watch in real-time as the neighboring cells geometrically expand to "swallow" the orphaned traffic.*

28.3 Flow Vector Fields (Weather Maps)

Traditional maps draw discrete lines between POPs, which become chaotic when mapping millions of "eyeball network" connections (individual consumer IPs). Inspired by meteorology, we can treat traffic direction and velocity as a continuous fluid dynamic system. By overlaying a grid on the geographic map, we compute the net directional flow vector for each grid cell and visualize it using animated streamlets or particle traces.

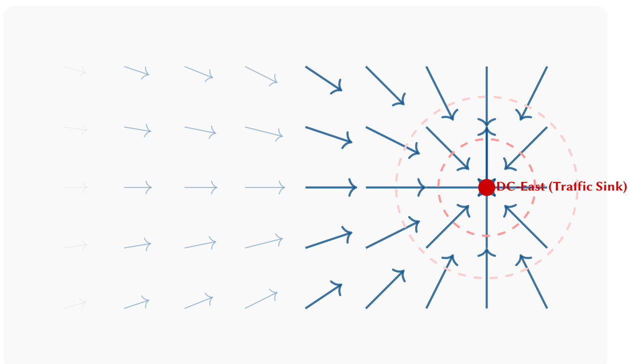


Figure 54. Flow Vector Field (Weather Map). By treating traffic as a continuous fluid, millions of individual eyeball connections are abstracted into a geographic "wind pattern." It is immediately obvious that traffic is heavily funneling into the DC-East sink.

Insight: *A Vector Field map turns network traffic into a "weather system." Operators can literally see a "hurricane" of traffic swirling towards a specific datacenter, or watch the "wind" reverse direction during a major BGP failover, grasping the macroscopic physics of the internet without looking at a single number.*

28.4 Poincaré Plots for Jitter and Variability

For latency-sensitive applications like Voice over IP (VoIP) or High-Frequency Trading, the *variability* of latency (jitter) is often more destructive than the absolute latency itself. Standard time-series lines hide jitter as dense "fuzz." A Poincaré plot (a type of return map) plots $Latency(t)$ on the X-axis and $Latency(t+1)$ on the Y-axis.

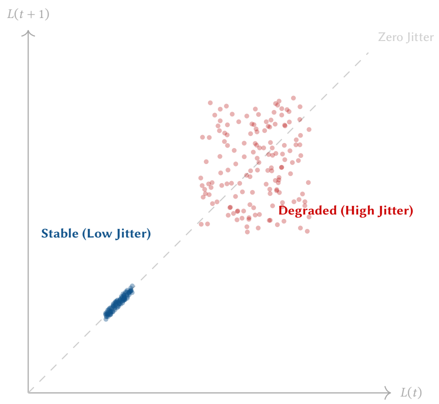


Figure 55. Poincaré Plot (Return Map) of Latency. The blue cluster shows a highly stable link (low jitter). The red cloud shows a link experiencing severe variance; the wide spread of points far off the dashed diagonal line geometrically quantifies the instability.

Insight:
A Poincaré plot turns the concept of "jitter" into a geometric shape (an attractor). A perfectly stable network is a single dot on the 45-degree diagonal line. As jitter increases, the dot expands into a "cloud." If the network is fluctuating between two distinct paths (e.g., a route flap), it forms distinct geometric clusters off the diagonal.

28.5 UMAP Health Landscapes (Manifold Learning)

A modern edge router generates hundreds of telemetry variables simultaneously (CPU, memory, temperature, optical Rx power, queue depths across 400 ports). We cannot plot 400 dimensions on a screen. Manifold learning algorithms like UMAP (Uniform Manifold Approximation and Projection) mathematically crush this hyper-dimensional state space down into a 2D topographic map.

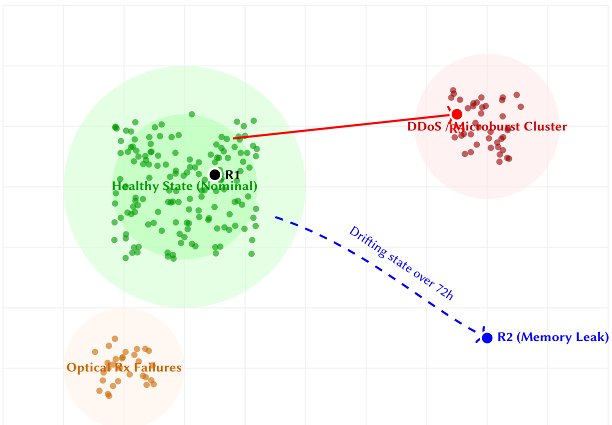


Figure 56. UMAP Health Landscape. The neural network learns the shape of historical telemetry and draws a 2D map. We can monitor the holistic, 400-dimensional health of R1, R2, and R3 simply by watching where their dots live on the geographic layout.

Insight:
UMAP creates a visual "Geography of Health." We plot all historical router states to form continents. A healthy router sits in the "Central Valley." As a memory leak slowly develops, the operator watches the router's dot physically migrate across the map, leaving the healthy continent and drifting into the "Memory Exhaustion Peninsula."

28.6 ML-Assisted Computer Vision on Telemetry Matrices

Borrowing from modern Convolutional Neural Networks (CNNs), we can treat a massive dense matrix of router interfaces (e.g., thousands of ports across a spine-leaf fabric) as an "image." By applying computer vision techniques, the ML model can identify highly localized, statistically anomalous regions.

Insight:
Instead of searching for a needle in a haystack, the dashboard draws a glowing bounding box around the needle. Just as a self-driving car highlights a pedestrian, the network dashboard draws an "Object Detection" contour around a cluster of ports experiencing a coordinated microburst.

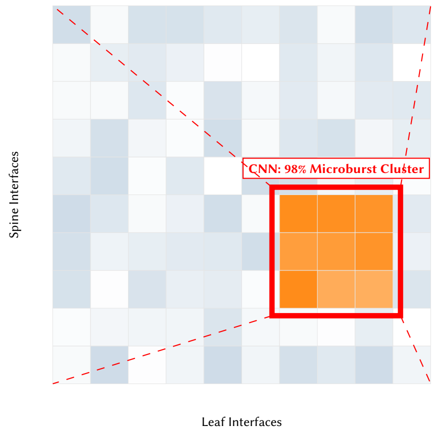


Figure 57. ML-Assisted Object Detection applied to a Telemetry Matrix. The algorithm treats the port metrics as an image, drawing a bounding box around a statistically significant cluster of anomalous interfaces, instantly drawing the operator's eye to the "object" of interest.

29 Control Plane, Policy, and Protocol Convergence

While the previous sections largely focused on the data plane (packet metrics, latency, queues), visualizing the *control plane*—the routing protocols and policies that dictate where packets should go—requires a different visual language. Control plane data is highly categorical, event-driven, and structural.

29.1 Protocol State Timelines (Gantt Charts)

When bringing up a new neighbor or diagnosing a flapping BGP session, observing a raw log of gRPC state change messages is cognitively taxing. A Protocol State Timeline maps categorical state machine transitions onto a continuous Gantt chart.

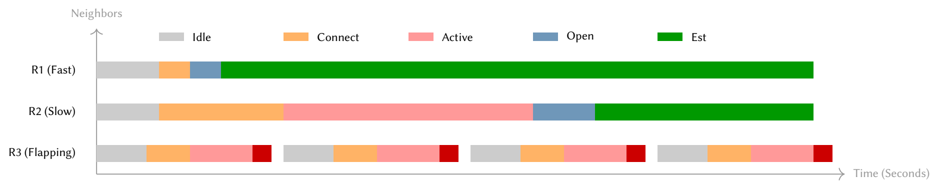


Figure 58. Protocol State Timeline (BGP). R1 converges perfectly. R2 experiences a delay in the Active state. R3 visually demonstrates a classic flapping scenario, looping through states without ever reaching Established, instantly indicating an authentication or parameter mismatch.

Insight:
A Protocol State Timeline visualizes the friction of convergence. If a router is misconfigured, the operator won't just see a "down" state; they will see a chaotic, repeating block of colors as the router repeatedly transitions from Connect → Active → OpenSent → Idle, visually revealing a protocol loop.

29.2 Convergence Ripple Maps

When a link fails, it triggers a cascade of control-plane updates (e.g., OSPF LSAs or BGP withdraws). Visualizing these updates as standard bar charts hides the geographic blast radius. A Convergence Ripple Map overlays propagating circular waves onto the topology.

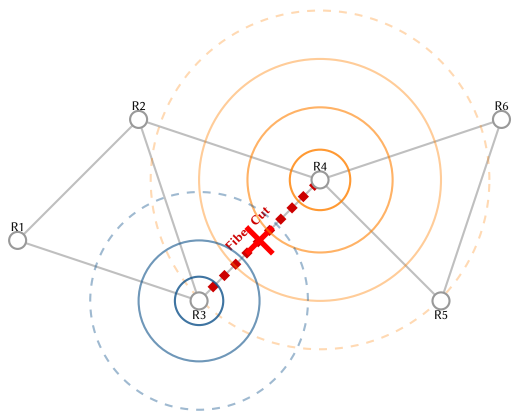


Figure 59. Convergence Ripple Map. The severed link between R3 and R4 generates control-plane updates. The expanding orange and blue rings visualize the propagation of these route withdrawals as they flood through the topology to neighbor nodes.

Insight:
A Ripple Map turns a router failure into a "stone dropped in a pond." Operators can watch the routing updates radiate outward. If the ripples hit a specific region and reflect backwards (a route loop) or get trapped, the topological nature of the convergence failure is immediately visible.

29.3 Reachability Matrices (Intent vs. Reality)

Network policy (what is *supposed* to happen) frequently diverges from network reality (what is *actually* happening) due to configuration drift or silent hardware failures. We can fuse these two datasets into an Adjacency Matrix where the columns are destinations, rows are sources, and the cells visualize the overlap.

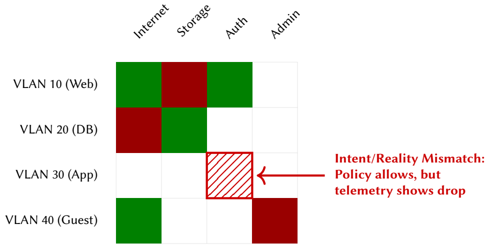


Figure 60. Intent vs. Reality Reachability Matrix. Solid green indicates policy and flow data agree (traffic is flowing). Solid red indicates agreement (traffic is blocked). The striped cell alerts the operator to a silent failure where the policy permits access, but packets are failing to reach the destination.

Insight:
By texturing the cells, we create a visual diff between policy and telemetry. A solid green cell means "Allowed and Flowing." A striped red/green cell instantly flags a violation: "The ACL policy allows this, but the IPFIX telemetry shows zero traffic is actually arriving."

30 Spatiotemporal Dynamics and Animation

The static encodings discussed thus far map time to the spatial X-axis (line charts, stream graphs) or to color intensity. However, network telemetry is fundamentally dynamic. Introducing *animation* allows us to reclaim the spatial axes for topological or structural mappings while delegating time to actual temporal playback.

Key spatiotemporal techniques include:

- **Animated Phase-Space:** The phase-space trajectories can be animated to show the "comet" orbiting the baseline knot, with the comet's tail length proportional to the rate of change (acceleration).
- **Flow Particles:** Rather than using static band thickness, flow volume and latency can be encoded as animated particles moving along links. Faster particles indicate low latency; denser particle clusters indicate high bandwidth.
- **Diurnal Breathing:** Playing back a 24-hour radial customer density tree reveals the geographic and topological "breathing" of the network as the sun rises and sets across different subnets.

Insight:
Animation exploits the human visual system's high sensitivity to motion. While a slow 5% increase in utilization might go unnoticed in a sparkline, the accelerating "pulse" of a congestion wave propagating across an animated topology commands instant attention.

30.1 Space-Time Prisms (Hovmöller Diagrams)

When analyzing a sequence of routers along a backbone path, operators need to see how a congestion wave propagates over both distance and time. A Space-Time Prism (or Hovmöller diagram) flattens the network topology onto the X-axis (e.g., Hop 1 to Hop N) and plots time continuously on the Y-axis.

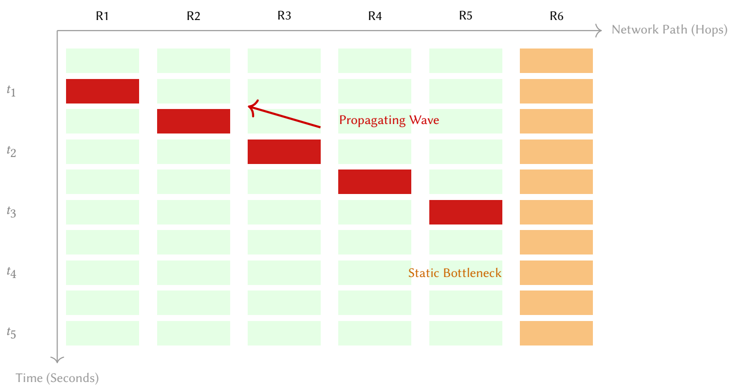


Figure 61. Space-Time Prism (Hovmöller Diagram). The X-axis represents physical routers along a path; the Y-axis represents time flowing downwards. R6 shows a static performance issue (vertical stripe). A congestion wave is seen originating at R1 at t_1 and propagating diagonally to R5 by t_3.

Insight:
In a Space-Time Prism, a static bottleneck appears as a vertical stripe. However, a routing cascade or propagating congestion wave appears as a diagonal streak, vividly illustrating the velocity of the incident as it travels through the network.

While static documents like this report cannot fully demonstrate animation, dashboard designers should consider motion an encoding channel of equal importance to position and color, provided it is used to encode data rather than merely for aesthetic flair.

30.2 Elastic Topologies (Warped Space-Time)

Geographic network maps plot routers at their physical coordinates (e.g., London and Sydney are far apart). However, network packets only care about *latency distance*. An Elastic Topology abandons physical coordinates and dynamically stretches or shrinks the map so that 1 centimeter = 10 milliseconds.

Insight:
If a 5ms intra-city fiber link suddenly fails and routing falls back to a 150ms satellite backup, the operator literally watches the city on the map "rip away" and float out into the ocean, providing a visceral, physical

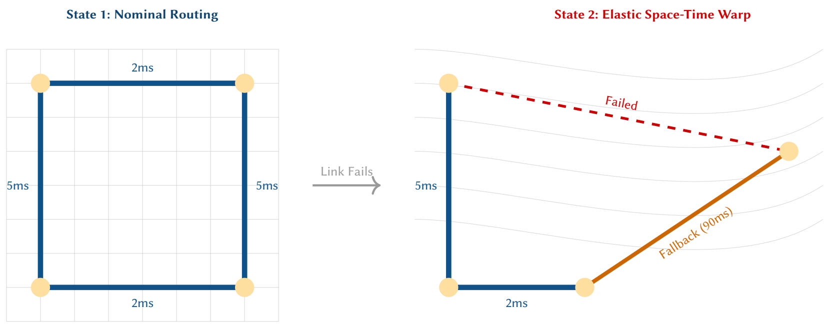


Figure 62. Elastic Topology (Warped Space-Time). In State 1, nodes are arranged symmetrically. In State 2, the primary 2ms link fails. Routing shifts to a 90ms backup path. The visualization dynamically warps the coordinate space, physically pushing the top-right node away to represent its new "latency distance" from the rest of the network.

30.3 3D Space-Time Cubes (Temporal Extrusion)

If a link goes down and comes back up while an operator is getting coffee, a standard 2D map shows a perfectly green network upon their return. To view the topology's history simultaneously, we extrude the 2D network map vertically along the Z-axis, turning Time into a physical dimension.

Insight: *In a Space-Time Cube, a router is not a dot; it is a vertical "pillar." A network link is not a line; it is a solid "wall." A flapping link instantly appears as a wall with "windows" or "bites" taken out of it, allowing an operator to scan 24 hours of topological stability in a single 3D glance.*

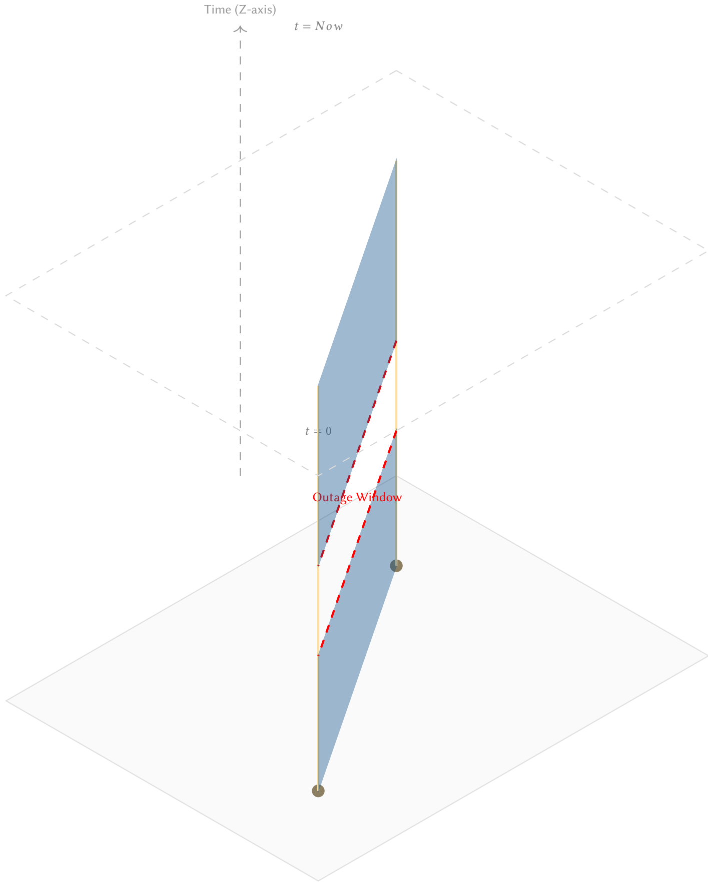


Figure 63. 3D Space-Time Cube. The network topology (two nodes and a link) is plotted on the floor, and time is extruded upwards. The link forms a continuous blue plane. The 1-hour outage is instantly visible as a physical "gap" or missing chunk in the wall.

30.4 BGP Contagion Replay (The Butterfly Effect)

Routing loops, prefix hijacks, and massive BGP route leaks propagate like biological viruses. When an operator is conducting a post-mortem, looking at a static graph of the final corrupted state is insufficient. A "DVR Replay" interface fuses a timeline scrubber with a series of small-multiple topological maps.

Insight: *By providing a DVR-style playback, operators can watch the "infection" spread. A BGP Replay interface visually proves the butterfly effect: a single AS (Autonomous System) incorrectly announcing a route turns red at t_1, infecting its peers at t_2, until the entire global map is bleeding traffic by t_4.*

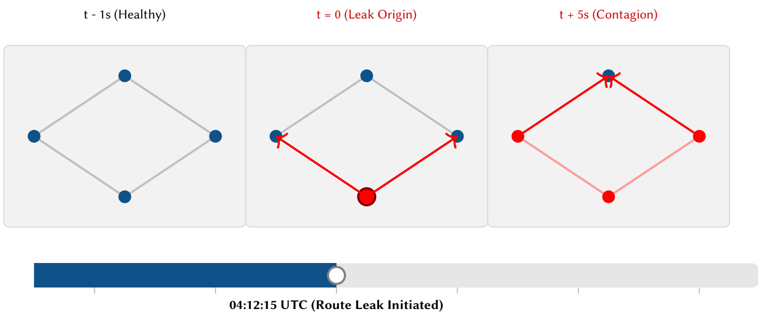


Figure 64. BGP Contagion DVR Replay. By dragging the playhead on the timeline, an operator watches the structural propagation of a routing error. What begins as a single misconfigured AS (t=0) rapidly spreads, poisoning neighbor routing tables and hijacking traffic in a visual cascade.

31 Implementation Strategy: Rendering Large Topologies

When building these advanced visualization systems in modern web browsers, the choice of rendering engine is critical. Browsers handle Document Object Model (DOM) updates poorly when the element count exceeds 1,000 nodes. We recommend the following strategies:

- **SVG vs. Canvas/WebGL:** Use Scalable Vector Graphics (SVG) (*Scalable Vector Graphics*) for highly interactive, smaller topologies (< 500 nodes) such as core and aggregation layers, because DOM elements attach cleanly to hover events and tooltips. However, for ultra-scale flow-level data, dense pixel displays, or hexbins ($> 10,000$ items), SVG will crash the browser. In these cases, render using HTML5 Canvas or WebGL, which pass the drawing instructions directly to the GPU.
- **WebWorkers:** Decouple the telemetry processing pipeline (e.g., Protobuf decoding, data aggregation, force-directed layout computation) from the main UI thread by using WebWorkers. This ensures the dashboard maintains a fluid 60 frames per second (fps) during interactions, even during massive state updates.

32 Conclusions and Recommendations

The visualizations surveyed in this report share a common thesis: *the right chart for the right question*. Line charts remain appropriate for single-metric trend analysis, but network operations routinely pose questions that demand richer encodings.

Design Decision 1: Visualization Selection Guide

- **Who talks to whom?** → Chord diagram (section 6)
- **Where does latency accumulate?** → Flame graph (section 11)
- **Which hop causes the delay?** → Waterfall chart (section 16)
- **Which interfaces are hot, when?** → Heat map (section 5)
- **Is the device healthy overall?** → Radar chart (section 8)
- **Which metrics correlate across devices?** → Parallel coordinates (section 14)
- **How is bandwidth distributed?** → Treemap (section 10)
- **How does traffic flow through tiers?** → Sankey (section 7)
- **What changed in the traffic mix?** → Stream graph (section 15)
- **What's the trend across 50 interfaces?** → Sparklines (section 9)
- **Which node is anomalous?** → Force-directed + highlight (section 13)
- **Where is the problem geographically?** → Geospatial overlay (section 17)
- **When did the micro-burst occur?** → Multi-resolution view (section 20)
- **What changed in the topology?** → Topology diff (section 21)
- **Is ECMP balanced?** → Path diversity view (section 22)

We recommend that NOC teams adopt a layered approach following Shneiderman's mantra [9]: begin with topology or geospatial overlays for spatial awareness (overview), add sparkline panels and heat maps for temporal density (zoom and filter), and introduce specialized views (chord, Sankey, flame, parallel coordinates) for targeted analysis (details on demand). The perceptual foundations in section 2 should guide encoding choices—use position for precision, color for pattern detection.

The anomaly detection feedback loop (section 19) represents a particularly important frontier: as ML-based detectors become standard in NOC operations, the visualization layer must support not just passive display but active sense-making—allowing operators to confirm, override, and label machine-generated alerts [33].

Telemetry collection has matured through gNMI and OpenTelemetry [31]. The rendering engines exist in D3.js [38] and Grafana [3]. The next frontier is *telemetry comprehension*—and that is fundamentally a visualization design problem [7].

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Revision History

Version	Date	Changes
1.1.0	March 2026	Added perceptual foundations, related work, and four new techniques
1.0.0	March 2026	Initial release